



Government of Tamil Nadu
State Planning Commission



REPORT

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Evaluation of Non-Communicable Disease Cascade Care under **Makkalai Thedi Maruthuvam in Tamil Nadu**

A CROSS-SECTIONAL STUDY



Report on
Evaluation of Non-Communicable Disease care cascade
under Makkalai Thedi Maruthuvam in Tamil Nadu
A Cross Sectional Study

State Planning Commission, Government of Tamil Nadu
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Foreword

'Makkalai Thedi Maruthuvam Programme' is an innovation in service delivery by the State. The programme is designed to ensure 100% scanning for Non-Communicable Diseases (NCD) and its follow up. The Dravidian Model Government has pioneered yet another welfare programme in the Country.

The State Planning Commission took up the impact Assessment of Makalai Thedi Maruthuvam scheme, a care cascade model in NCD management. The study was undertaken in association with the institute of Community Medicine, Madras Medical College along with the Community Medicine Departments across all the Medical Colleges in Tamil Nadu.

The study indicates that Out-of-pocket medical expenditure for the lower income groups has halved after the scheme was implemented. Another important finding of the survey is that before MTM, only one third of the people from the low income groups were screened for diabetes and hypertension and this rose to nearly half. Many such insights have emerged from this survey.

I thank the study team of Community Medicine Department of Madras Medical College and other Medical Colleges across districts for their commendable work in conduction the Study.

I assure that the State Planning Commission will continue in its endeavor to document the impacts of such interventions and evaluate the flagship programmes of the State Government.

Vice Chairman,
State Planning Commission.

This report was jointly prepared by Institute of Community Medicine, Madras Medical College and State Planning Commission, Government of Tamil Nadu. The study was funded by State Planning Commission, Government of Tamil Nadu.

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We thank the Vice-Chairman, State Planning Commission, for entrusting the work and guiding the work from time to time. We thank Dr J.Amalorpavanathan, Member, State Planning Commission and Dr G Sivaraman for their guidance in designing the study. We thank the Full Time Member, Additional Fulltime Member, SPC for their constant support and guidance in this study. We also thank the Member-Secretary, SPC for the facilitation and coordination in conducting the study.

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We thank the CRMIs, Nodal officers across the Institute of Community Medicine Departments of all Medical Colleges in efficient data collection and on time completion of the Study.

We thank the core committee members from the select Medical Colleges for designing the survey tool, translation and conversion into online data collection tool on web-based survey tool which helped for faster compilation.

We thank all the Staff from Health and Social Welfare Division, State Planning Commission for their constant support.

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Executive Summary

“**Makkalai Thedi Maruthuvam**” (MTM) scheme is a flagship program of the Government of Tamil Nadu offering a holistic and comprehensive set of “**Home Based Health Care Services**” to ensure a continuum of care, sustainability of the services, and meet the health needs of beneficiaries in the family. State Planning Commission, as part of its evaluation of various flagship schemes of the Government, proposed to evaluate the cascade of care delivered through the MTM scheme.

Institute of Community Medicine, Madras Medical College, Chennai was given the responsibility of conducting the survey in coordination with the Community Medicine department of other Government Medical Colleges of Tamil Nadu. In this regard, a core **Committee** involving faculty from 8 different Medical Colleges was formed to formulate the protocol and develop the questionnaire. The questionnaire was piloted, and necessary corrections were made.

The objectives of the survey were to find the following:

- Proportion of eligible people ever visited by a Women Health Volunteer
- Proportion of eligible population screened for Diabetes and HTN in the last one year
- Proportion of eligible population under treatment among DM/HTN patients
- Proportion of eligible population under follow up among DM/HTN patients
- Proportion of eligible population under control for DM/HTN
- Proportion of eligible population covered under Palliative care (among who required)
- Proportion of eligible population covered under Dialysis
- Proportion of women screened for CA Cervix
- Proportion of women screened for CA Breast

Methodology

A state-wide cross-sectional survey is conducted to obtain representative estimates from the adult population, aged **>30 years**, in all districts of Tamil Nadu. All individuals aged > 30 years of age and residing in the particular address for more than 6 months were included. The minimum sample size required is **5760 households**. Multi-stage cluster sampling method was used for selecting the required number of samples. The term cluster refers to Villages in rural areas and streets in urban areas.

In each cluster 30 households were randomly selected and all eligible members present in the household were **interviewed** after obtaining informed written consent.

The number of clusters in each district (urban and rural) was selected proportionately to the population size. The total number of clusters selected across the state was 197 (after making decimal corrections), out of which 110 were rural and 87 were urban clusters. The number and list of clusters in each district are given in Annexure 1. The clusters were selected following a multistage random sampling method by the core committee.

Key Results

Among the study participants, 4155 (60%) were from rural areas and 2701 (40%) belonged to urban localities. Gender distribution was 2575 (37.5%) men, and 4279(62.4%) were women. Based on Community, 24.8% did not want to tell their community. Among those who revealed, 825(16%) belonged to the SC community and 220 (4.26%) belonged to the ST community.

Almost 78.84% of the study participants were aware of MTM and 73 % were ever visited by WHV and 65.8% were briefed about MTM services by WHV. While there was no gender difference, the awareness on MTM and proportion of people visited by a WHV was significantly lower in urban areas compared to rural areas. Similarly, awareness on MTM was significantly higher among the vulnerable category ST compared to non-SC / ST. Significantly higher proportion of SC and ST community people were visited by a WHV and were briefed about MTM services compared to non-SC/ST people.

Screening for Hypertension: Almost 81.25% of the whole study population have ever screened for Hypertension, among them 93.8% had screen themselves for Hypertension in the last year. Among those screened in the last year, 3/4th was screened through MTM specifically almost 50% were screened by field workers at doorstep under MTM. A **higher proportion of women were screened compared to men** (82.49% vs 79%). Screening through MTM in the last year was also higher among women (70.6% vs 58%). Under MTM, screening by field workers at the doorstep was high among women compared to men (51% vs 40%)

There was an urban-rural difference in hypertension screening coverage, with a higher coverage rate in rural areas (82.4%) compared to urban areas (79.3%). There was a significantly lower screening coverage through MTM for Hypertension in urban areas (53.9% urban vs 78.5% rural). Only 36% were screened by field workers in urban localities compared to 57% in rural areas.

Based on caste category there was no significant difference in overall screening coverage. However, **screening coverage through MTM services was significantly higher among SC**

and ST categories compared to non-SC/ST categories. Specifically, the ST population (79.4%) had a higher proportion screened through MTM field workers compared to SC and non-SC/ST population (53.8% & 49.1%). MTM screening through institutions was significantly higher among SC (23%) and lowest among ST (7.4%) compared to non-SC/ST (17.8%).

Proportion of Hypertension

Among the study population, 22% were hypertensive. There was no gender difference, but urban population had a higher proportion of **hypertensives** compared to rural areas (25% vs 20%). People belonging to non-SC/ST had a higher proportion of HTN compared to SC (23% vs 18.3%). Among the hypertensives, 1/5th were diagnosed in the last 1 year and 96% reported to be on treatment. There was no significant difference in the treatment coverage based on gender, caste category and locality. Among the hypertensives, 2/3rd were receiving treatment under MTM and specifically, 41% of the Hypertensives had received treatment at their doorsteps through field workers. There was a significant difference in **treatment** received through MTM based on locality. Only 48% of hypertensives in urban areas received treatment through MTM, compared to 74% in rural areas. Similarly, only 27.7% of hypertensives in urban areas had been dispensed drugs through field workers compared to 52% in rural areas. Among different caste category, 3/4th of the hypertensives belonging to SC and ST had received treatment through MTM compared to only 60% among non-SC/ST category.

Follow up and Control

Among the hypertensives, 90% reported compliance to drugs in the last 1 week and 70% had recorded their blood pressure at least once in the past 3 months. Based on the latest recorded BP within the last 3 months, 35.4% of the hypertensives had their BP under control. There was no gender, caste and locality-based difference.

Diabetes Mellitus

Among the study participants, 80% had screened for DM. Among those screened, 93% had their screening done in the last one year. **Sixty seven percent of those screened in last one year, were screened through MTM.** Specifically, 47% were **screened** by field workers at their doorstep and 21% through institutions. **A higher proportion of women had screened for DM compared to men (81% vs 77%).** There was no urban rural difference. People belonging to ST category had a significantly higher proportion screened for DM compared to SC and non SC/ST categories.

Among those screened through MTM in the last one year, there was a significant gender difference with preponderance towards women. Similarly, a higher proportion of women(50%) were visited by field workers under MTM compared to men(43%).

Rural counterparts (77%) had a higher proportion screened for DM through MTM compared to urban (53%) which was also reflected in screening through field workers (54.5% vs 35.3%) and institutional screening (22.4% vs 17.5%).

Though there was no significant difference between caste category on number screened for DM , there was a gradient based on caste category, w.r.t **screening** for DM under MTM . Eighty five percent of those among ST were screened through MTM , compared to 80% in SC and 66% among non -SC/ST category.

Such difference was also observed in proportion screened through field workers with the highest reported among ST category 72.6%.Institutional screening for DM under MTM was highest among SC (18.22%) and the lowest among ST community.(12.29%)

The overall proportion of people who reported to be a diabetic was 21% , with men reporting higher compared to women (22.5% vs 19.4%).Similarly urban had higher proportion of DM reported compared to rural (25.1% versus 17.6%). There was **caste** difference in the DM proportion with highest among non-SC/ST (22%)and lowest in ST(9.55%).

Among those diagnosed with DM, 16.6% were diagnosed in the last one year and 97% reported to be on treatment. Fifty five percent of those with DM received treatment under MTM and 32% got their Medicines dispensed through Field workers. A significantly higher proportion of rural population (74.5%) with DM were covered under treatment by MTM compared to urban (48.08%). Similarly 52.3% of those with DM in rural areas were dispensed drugs by field workers whereas only 27.7% in urban.

With regard to caste category, non SC/ST had a significantly lower proportion receiving treatment through MTM compared to SC and ST. Similarly, 70% of ST people with DM were receiving drugs dispensed through Field workers.

Follow up and control

Among the diabetic, 66% had their blood sugars tested in last 3 months and 30% of the diabetics had their sugar levels under control. There was no **difference** based on gender, caste and locality.

Cancer screening

Cancer screening uptake was lower overall. Only 4%, had ever undergone screening for oral cancer. For cervical 14% of women had ever undergone **screening** for cervical cancer and breast cancer. There was no rural-urban difference and gender difference (oral cancer).

However, ST had the lowest screening coverage for oral **cancer** and cervical cancer, compared to other category.

Counselling services

Under MTM, counselling for cancer screening should be provided as a service. However, only 17%, 35% and 39% had ever received counselling regarding screening for oral, cervix and breast cancer respectively. While there was no difference in this proportion, based on locality, difference based on Caste category was obvious. Counselling services were reported to be lowest among ST category compared to others.

Conclusion

The results show that MTM has reached out to the most vulnerable groups like women, rural and SC/ST category. Among the various services offered, diabetes and hypertension screening and treatment coverage has reached the most. This indicates, de-professionalization of health care by involving Women Health Volunteers has made a major impact in reaching out to the most vulnerable population. While the MTM is enabling to provide of universal coverage for NCD, the thrust areas include reaching out to the urban areas and devising strategies for improving overall cancer screening.

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Definitions

Hypertension diagnosis: Ever told to have hypertension and /or on medication.

Hypertension Treatment: Taking medication over the last two weeks.

Hypertension Treatment through MTM: Getting medication either from Women Health Volunteer or any government health facilities of the state.

Hypertension Control: SBP <140 mmHg and DBP <90mmHg in any last recorded value in the past 3 months

Diabetes Diagnosis: Ever told to have diabetes and /or on medication.

Diabetes Treatment: Taking medication over the last two weeks.

Diabetes Treatment through MTM: Getting medication either from Women Health Volunteer or any government health facilities of the state.

Glycemic Control: Fasting Blood Glucose ≤ 120 mg/dl and PPBS ≤ 180 mg/dl in any last recorded value in the past 3 months.

Screening for oral cancer: Clinical oral examination done ever in both men and women by a healthcare professional for early signs of oral cancer.

Screening for breast cancer: Clinical breast examination done ever for women ≥ 30 years of age by a healthcare professional for breast cancer.

Screening for cervical cancer: Screening tests ever done for cervical cancer for women aged ≥ 30 years, by means of either/and Visual Inspection with Acetic Acid (VIA), pap smear

Key Findings of the study

73% of the survey participants had been benefitted by the MTM door-to-door services

Diabetes Screening and treatment

2/3rd of newly diagnosed DM detected through MTM in the last 1 year.

More than half of Diabetics receiving treatment through MTM.

One -third of Diabetics received treatment through MTM Field functionaries.

10% of Diabetics had good glycemic control.

Hypertension Screening and treatment

3/4th of newly diagnosed HTN detected through MTM in the last 1 year.

63% of hypertensives received treatment through MTM.

45% of hypertensives received treatment through MTM field functionaries.

35% of hypertensives had good blood pressure control.

Cancer Screening

3.7% ever screened for oral cancer

11.06 % of women ever screened for cervical cancer

14.2 % of women ever screened for breast cancer

Other services

40% of those requiring palliative care covered through MTM -field staff.

48% of those requiring physiotherapy services covered through MTM -field staff.

52% of those requiring dialysis services covered through MTM -field staff.

EVALUATION OF MAKKALAI THEDI MARITHUVAM

Introduction

"Makkalai Thedi Maruthuvam" (MTM) scheme is a flagship programme of the Tamil Nadu government that provides a holistic and comprehensive set of "Home Based Health Care Services" to ensure a continuum of care, sustainability of the services, and meeting the health needs of beneficiaries. "MTM" scheme is conceptualized in such a way that a field-level team would provide home-based health care services especially targeting non-communicable diseases (NCD) which includes preventive and promotive services like screening for hypertension, diabetes and providing counselling regarding lifestyle modification and cancer screening, and curative services for line-listed beneficiaries such as the delivery of Hypertension/Diabetes Mellitus drugs for patients who are 45 years and above and those with restricted mobility, Home-based Palliative Care and Physiotherapy services, caring for End Stage Kidney Failure patients, referral for Essential Services, identification of children with congenital problems or any other health needs in the family which needs to be informed and followed up. The existing NCD services provided at Public Health Facilities in the State also come under the umbrella of MTM. Palliative care, Physiotherapy, and Continuous Ambulatory Peritoneal Dialysis services at Institutions are provided under the MTM scheme, and Patients eligible for Home-based MTM services are referred to their respective PHC.

MTM field team includes Women Health Volunteers (WHV), Mid-Level Health Providers (MLHP), Village Health Nurses (VHN), Health Inspectors (HI), Palliative Care Staff Nurses, and Physiotherapists with the monitoring support of other public health field staff. The institutional level MTM team includes MTM - NCD Staff Nurses at DPH, DMS & DME institutions, and Institutional Nodal Officers at DME & DMS institutions for providing comprehensive NCD services at primary, secondary, and tertiary care levels.

The comprehensive range of NCD care services that are delivered at the doorstep under the scheme MTM is as follows:

1. Population-based screening for those aged 18 years and above for 10 common conditions including Diabetes, Hypertension, Cervical Cancer, Breast Cancer, and Oral Cancer.
 - a. Screening for Diabetes and Hypertension for all men and women >30 years

- b. Screening for CA cervix and breast for all women > 30 years
 - c. Counselling on lifestyle management for all adults
- 2. Deliver HTN/Diabetes drugs to registered patients aged 45 and above & to those with restricted or poor mobility.
- 3. WHV motivates women for cancer screening.
- 4. Home-based palliative care services by the Palliative Care Staff Nurse for patients with chronic debilitating illnesses who have difficulty visiting health facilities.
- 5. Home-based Physiotherapy services by physiotherapists for elderly, home-bound patients and those with restricted mobility.
- 6. Home-based outreach services by delivering peritoneal dialysis bags to patients under CAPD by the Palliative Care Staff Nurse.

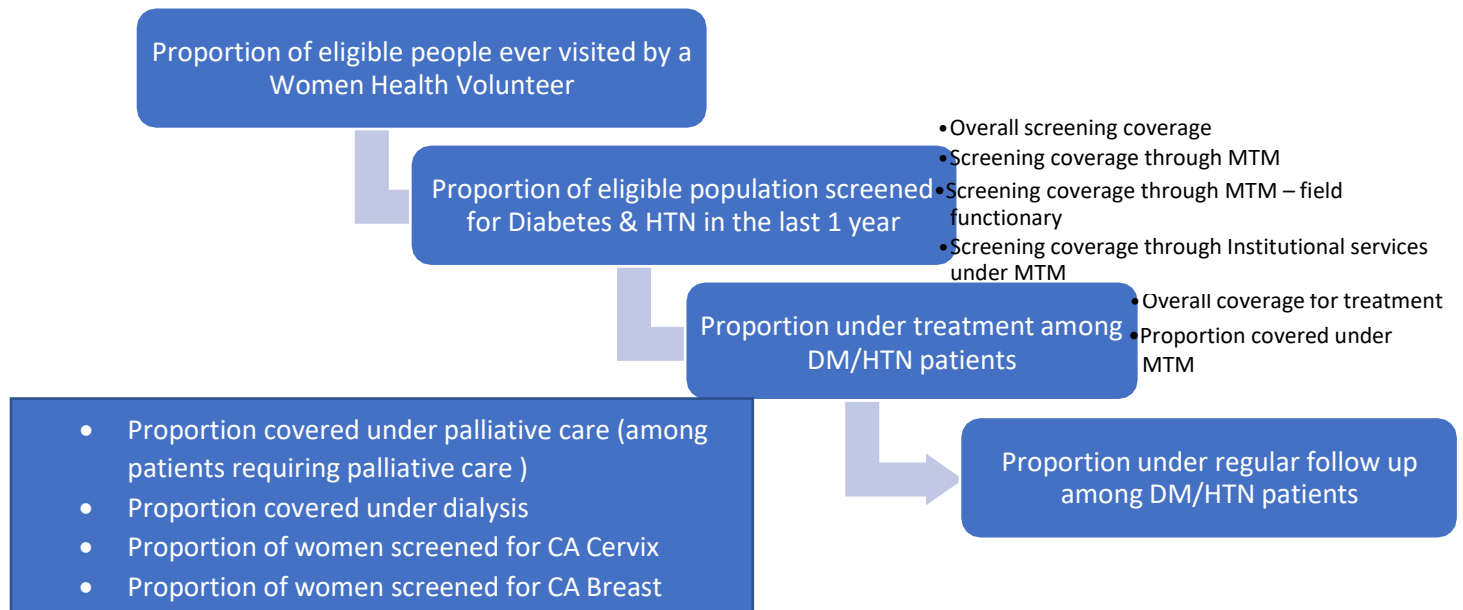
The scheme was launched in a phased manner. Phase one was launched in August 2021 covering 50 Universal Health Coverage blocks and one zone in 3 corporations (Tirunelveli, Coimbatore, and Greater Chennai Corporation). Later the scheme was expanded to cover the entire state by September 2021. The scheme covers both urban and rural areas.

It is more than a year since the expansion of MTM services to cover the entire state. Hence, it is essential to understand the coverage of the services rendered by the MTM scheme in Tamil Nadu. Therefore, this study is mandated by the State Planning Commission to evaluate the coverage of the services under MTM scheme in Tamil Nadu

Aim of the survey

To evaluate the reach of the cascade of services for non-communicable diseases rendered under Makkalai Thedi Maruthuvam in Tamil Nadu

Specific Objectives



The study also compared the above-mentioned indicators across gender, caste, and urban-rural locality. The coverage of the services is given for urban-rural areas in the state per gender for SC/ non-SC community group.

Methodology

Study design: Cross-sectional study

A state-wide cross-sectional survey was conducted to obtain representative estimates from the adult population, aged **>30 years**, in all districts of Tamil Nadu.

Study Population: All adults aged 30 years and above in Tamil Nadu was included in the survey.

Inclusion criteria

All individuals aged > 30 years of age and residing in the address for more than 6 months were included.

Exclusion criteria

Individuals who did not consent to participate in the study were excluded.

Sample Size

The number of the sample size required to be included in the sample with 95% confidence was calculated using the following formula and assumptions.

$$N = \frac{Z\alpha^2 pq}{d^2}$$

Wherein,

$Z\alpha$ - Level of confidence (for $\alpha=0.05$ and 95% confidence level) = 1.96

p - Estimated baseline levels of the indicators = 0.50

q - (1- p)

d - Margin of error = 0.05 $n = 1.96 \times 1.96 \{0.50 (1-0.50)\} / 0.05 \times 0.05$; $n = 384$

Design effect (Deff) = 1.5;

Number of community -sex-urban/rural estimates = 8

Corrected Sample Size		
Sample Size for estimating coverage at the state level	estimated sample size x Design effect x number of community -sex-urban/rural estimates / anticipated response rate	$384 \times 1.5 \times 8 / .8$ = 5760

The minimum sample size required is 5760 households.

Sampling Method

Multi-stage cluster sampling method was used for selecting the required number of samples. The term cluster in this manual is synonymous with the final survey unit i.e., Villages in rural areas and streets in urban areas. The cluster size was fixed as 30 households. The sample was drawn from all districts based on population proportion to size. Based on population proportion to size, no of clusters from each district was allocated. Then within each district, it was further stratified into urban and rural areas.

In urban areas, a three-stage cluster sampling method was adopted. In the first stage, Wards were selected by simple random sampling method. In the second stage, one street from the selected ward was chosen by a simple random sampling method. The sampling frame of the selected street was prepared by mapping and line listing of the households. Then, households were selected by circular systematic random sampling method. From the selected household, all eligible adult individuals were selected for the study.

In rural areas, a three-stage cluster sampling method was adopted. In the first stage, blocks were randomly selected. In the second stage, one village in the selected block was selected by simple random sampling method. Each village was considered as a cluster. The sampling frame of the selected village was prepared by mapping and line listing of the households. Then, households were selected by circular systematic random sampling method. From the selected household, all eligible adult individuals were selected for the study.

The procedure of mapping and line listing of households is described in Annexures (1&2).

Sample Allocation:

The total sample size is 5,760 households. With a cluster size of 30 in each, to cover the 5760-sample size the total number of clusters required is 197. This number is considered adequate to give an adequate representation of the state. The number of clusters from each district is determined based on the population proportion to size. Within each district, the share of the urban and rural clusters is determined based on population proportion. The illustration of cluster allocation is shown in Table 1. From each cluster selected, 30 households will be selected.

Table 1. Number of clusters selected from each district.

S.No	Districts	Total Population	Proportion of total population	Rural Clusters	Urban Cluster	Total clusters
1	Ariyalur	821903	1.04	2	1	3
2	Chengalpattu	2495264	3.16	3	3	6
3	Chennai	7564573	9.58	0	18	18
4	Coimbatore	3802600	4.81	3	6	9
5	Cuddalore	2845659	3.6	5	2	7
6	Dharmapuri	1640111	2.08	3	1	4
7	Dindigul	2357453	2.98	4	2	6
8	Erode	2462682	3.12	3	3	6
9	Kallakurichi	1515556	1.92	3	1	4
10	Kancheepuram	991911	1.26	2	1	3
11	Karur	1163192	1.47	2	1	3
12	Krishnagiri	2050494	2.6	4	1	5
13	Madurai	3339871	4.23	4	4	8
14	Mayiladuthurai	998251	1.26	2	1	3
15	Nagapattinam	764404	0.97	1	1	2
16	Nagercoil	2047073	2.59	2	3	5
17	Namakkal	1886197	2.39	3	2	5
18	Perambalur	615452	0.78	1	1	2
19	Pudukkottai	1763947	2.23	4	1	5
20	Ramanathapuram	1477839	1.87	3	1	4
21	Ranipet	1310551	1.66	2	1	3
22	Salem	3812709	4.83	5	4	9
23	Sivaganga	1460468	1.85	3	1	4
24	Tenkasi	1556828	1.97	2	2	4
25	Thanjavur	2627027	3.33	4	2	6
26	The Nilgiris	805062	1.02	1	1	2
27	Theni	1362278	1.72	2	2	4
28	Thiruvallur	2580046	3.27	4	2	6
29	Thiruvavarur	1378034	1.74	3	1	4
30	Thoothukkudi	1916097	2.43	3	2	5
31	Tiruchirappalli	2983413	3.78	4	3	7
32	Tirunelveli	1806957	2.29	2	2	4
33	Tirupathur	1290884	1.63	2	1	3
34	Tiruppur	2722984	3.45	4	3	7
35	Tiruvannamalai	2686778	3.4	5	1	6
36	Vellore	1705682	2.16	2	2	4
37	Villupuram	2248886	2.85	5	1	6
38	Virudhunagar	2129709	2.7	3	2	5
		78988825	100	110	87	197

Survey Teams

Almost all districts had a medical college with interns posted in Community Medicine and they were allotted the selected clusters in the district for data collection. Those in close vicinity were assigned to districts where no medical colleges existed. A faculty member in the department of Community Medicine was identified as the Nodal officer and interns posted in Community Medicine in the month of February- March 2023 were part of the team for data collection. Each survey team had 2 Medical Interns (one male and one female CRMI fluent in Tamil). The teams and the Nodal officer were trained for all survey-related work like cluster boundary mapping, household enumeration, household listing, data entry on handheld devices, interview tool use, etc.

Study Tool

A semi-structured interviewer-administered questionnaire was prepared in Tamil. English translation was done. The questionnaire included the following sections:

1. Demographic details
2. Diabetes Mellitus – screening, treatment and follow up
3. Hypertension – screening, treatment and follow up
4. Oral Cancers – screening and referral
5. CA cervix – screening and referral (only for adult women)
6. CA breast – screening and referral (only for adult women)
7. Home-based palliative care -only for those eligible for palliative care
8. Home-based dialysis – only for patients requiring dialysis

USE OF HANDHELDS FOR DATA COLLECTION

An android based mobile phone was used to collect survey data making the process easy, quick and flexible. Epicollect5 software was used to collect the data.

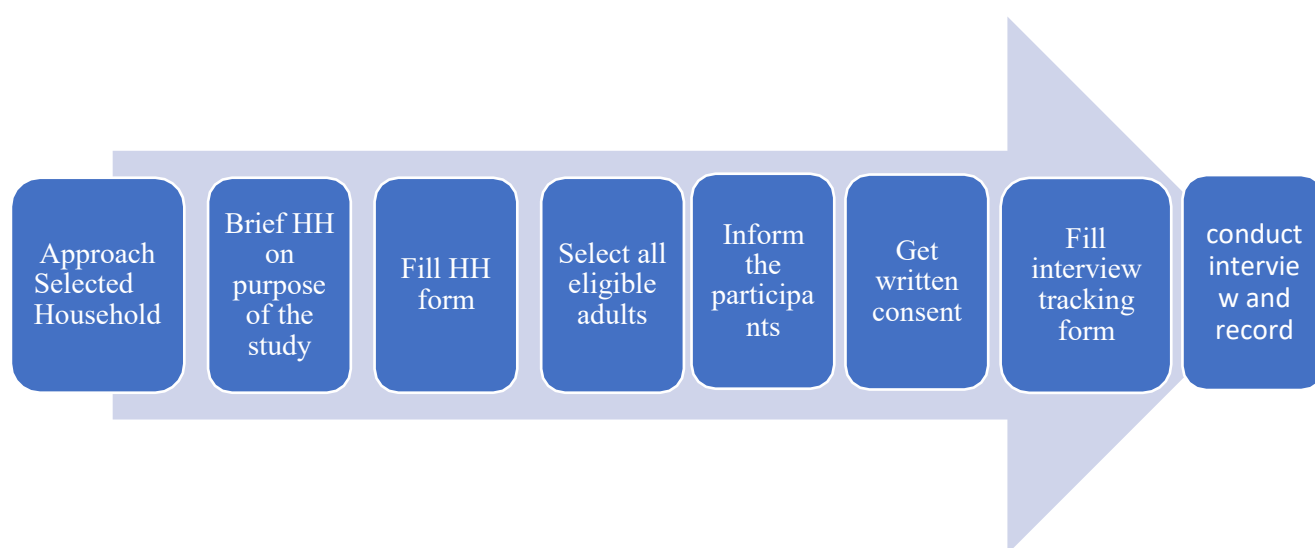
The Epicollect5 app was pre-tested on handheld devices.

Microplanning of the survey

Process in the field



Process in the household:



Data analysis

The process for creating the final dataset comprised of downloading the data and cleaning the data. SPSS version 21 was used for data analysis.

Completing the Data Book

The Data Book is a full tabulation, by gender - community - sex, of the data from all the questions in the study tool. It is intended to serve as the basis for the final report, to guide on what results to include and highlight in the report.

Each of the data tables contain results for urban-rural locality for both sexes and for each sex separately. For each group in the table, the point estimate is given (proportion or mean) along with 95% confidence interval (except for the demographic information) and the “n” (the total number of individuals included in the analysis for that group).

Ethical considerations:

The survey protocol was subjected to the Directorate of Public Health and Preventive Medicine Ethics Committee and Scientific Committee for their thorough review and approval. Informed consent was obtained from every survey participant before conducting the interviews. The participants were given information on the services rendered by MTM.

Results

Among the households approached, 5305 households were available for data collection.

Figure 1. No of eligible participants who participated in the survey

Number of eligible participants available in households = 6963

Number of eligible participants who gave consent and were interviewed = 6856

The data collection was started on 27/03/2023 and completed in 13/04/2023.

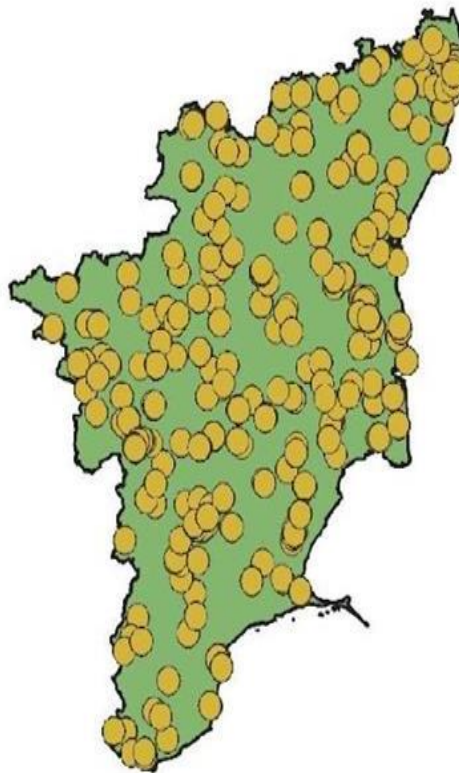


Figure 2. Geospatial distribution of the study participants

The demographic profile of the study participants is given in table 2. The distribution of the study participants is like the demographic structure of Tamil Nadu. As per the estimated midyear population, the urban rural ratio is represented in this study. According to Census 2011, the gender split is 50/50, although in this study there were more women participants than men.

Table 2. Demographic profile of the study participants

Demographic variable		Frequency	Percentage
Mean Age (SD) in years		51.9	13.9 years
Age group (n-6856)	30-40 years	1830	26.7
	41-50 years	1625	23.7
	51-60 years	1529	22.3
	61-70 years	1223	17.8
	>70 years	649	9.5
Gender (n-6856)	Men	2575	37.5
	Women	4279	62.4
	Third gender	2	0.3
Locality (n-6856)	Rural	4155	60.6
	Urban	2701	39.4
community groups (n-6856)	Other Communities (OC)	106	1.5
	Backward Class (BC)	2416	35.2
	Most Backward Class(MBC)	1589	23.2
	Scheduled Caste(SC)	825	12.0
	Scheduled Tribes(ST)	220	3.2
	Not willing to reveal	1700	24.8
Religion (n-6856)	Hindu	6086	88.8
	Muslim	254	3.7
	Christian	357	5.2
	Others	7	0.1
	Not willing to reveal	152	2.2
Economic class* (n-6856)	Class I (Upper Class)	839	12.2
	Class II (Upper Middle Class)	1336	19.5
	Class III (Middle Class)	2035	29.7
	Class IV (Lower Middle Class)	1457	21.3
	Class V (Lower Class)	1189	17.3
Education status (n-6856)	Illiterate	1876	27.4
	Primary Education	1321	19.3
	Middle School Education	1803	26.3
	High School Education	977	14.3
	Diploma	200	2.9
	Graduate	679	9.9
Occupation status (Rural; n-4151)	Self-employed in agriculture	869	20.9
	Self-employed in non-agriculture	294	7.1
	Regular Wage/ Salaried Employee	275	6.6
	Casual Labor in agriculture	375	9.0
	Casual Labor in non-agriculture	416	10.0
	Others	436	10.5
	Housewife	876	21.1
	Unemployed	610	14.7
Occupation status (Urban; n – 2680)	Self -employed	474	17.7
	Regular Wage/Salaried Employee	323	12.1
	Casual Labor	296	11.0
	Others	179	6.7
	Housewife	902	33.7
	Unemployed	506	18.9

*- B G Prasad Classification

Table 3. Proportion of people who received services under Makkalai Thedi Maruthuvam

Services	Frequency	Percentage	(95% CI)
Proportion Aware of MTM (n-6856)	5405	78.84%	(77.8-79.8)
Proportion ever visited by MTM field functionary (WHV) (n-6856)	4988	72.75%	(71.6-73.8)
Proportion briefed about MTM by WHV(n-4988)	4513	90.48%	(89.6-91.2)
Services related to Diabetes Mellitus			
Proportion ever screened for DM (n-6856)	5448	79.47%	(78.5-80.4)
Proportion screened for DM in the last 1 year (n- 6856)	5065	73.88%	(72.8-74.9)
Proportion screened for DM through MTM (n-5448)	3685	67.64%	(66.3-68.8)
Proportion screened for DM through MTM by field functionaries (n-3685)	2566	69.63%	(68.1-71.1)
Proportion screened for DM through MTM institutions. (n-3685)	1119	30.37%	(28.8-31.8)
Proportion ever diagnosed with DM (n-6856)	1419	20.70%	(19.7-21.6)
New diagnosis for DM in the last one year (n-1419)	236	16.60%	(14.7-18.6)
New diagnosis for DM in the last one year through MTM (n-236)	153	64.80%	(58.3-70.9)
Proportion on treatment for DM (n-1419)	1382	97.39%	(96.4-98.1)
Proportion on treatment for DM through MTM (n-1419)	774	54.55%	(51.9-57.1)
Proportion of DM dispensed medicines through WHV for DM (n-1419)	511	36.0%	(33.5-38.6)
Proportion of DM who have checked their blood glucose level in the last 3 months(n-1419)	940	66.24%	(63.7-68.7)
Proportion of DM with good glycemic control (n-1419)	139	9.8%	(8.3-11.4)
Proportion of DM patients who shifted from private to public (n-331)	97	29.3%	(24.4-34.5)
Services related to Hypertension			
Proportion ever screened for HTN (n-6856)	5570	81.25%	(80.3-82.1)
Proportion screened for HTN in the last 1 year (n- 6856)	5225	76.21%	(75.1-77.2)
Proportion screened for HTN through MTM (n-5225)	3847	73.6%	(72.4-74.8)
Proportion screened for HTN through MTM by field functionaries (n-3847)	2732	71.02%	(69.5-72.4)
Proportion screened for HTN through MTM institutions. (n-3847)	1115	28.98%	(27.5-30.4)
Proportion ever diagnosed with HTN (n-6856)	1511	22.04%	(21.0-23.0)
New diagnosis for HTN in the last one year (n-1511)	294	19.46%	(17.4-21.5)
New diagnosis for HTN in the last one year through MTM (n-289)	218	75.43%	(70.0-80.2)
Proportion on treatment for HTN (n-1511)	1454	96.2%	(95.1-97.1)
Proportion on treatment for HTN through MTM (n-1511)	947	62.67%	(60.1-65.1)
Proportion of HTN dispensed medicines through WHV (n-1511)	684	45.30%	(42.7-47.8)
Proportion of HTN who have checked their blood pressure level in the last 3 months (n-1511)	1050	69.49%	(67.1-71.8)
Proportion of HTN with good blood pressure control (n-1511)	535	35.41%	(32.9-37.8)
Proportion of HTN patients who shifted from private to public (n-292)	77	26.4%	(21.4-32.0)
Services related to Cancer			
Proportion advised to get Oral Cancer Screening (n-6856)	1168	17.04%	(16.1-17.9)
Proportion ever screened for Oral Cancer (n-6856)	265	3.87%	(3.4-4.3)
Proportion advised to get Cervical Cancer Screening (n-4058)	1433	35.31%	(33.8-36.8)
Proportion ever had Cervical Cancer Screening (n-4058)	449	11.06%	(10.1-12.0)
Proportion advised to get Breast Cancer Screening (n-4058)	1600	39.43%	(37.9-40.9)
Proportion ever had Breast Cancer Screening (n-4058)	578	14.24%	(13.1-15.3)
Other services			
Proportion covered under Palliative Care through MTM (n-319)	205	64.26%	(58.7-69.5)
Proportion covered under Palliative Care through MTM-field staff (n-319)	126	39.50%	(34.1-45.1)
Proportion receiving Physiotherapy services through MTM (n-320)	213	66.56%	(61.1-71.7)
Proportion receiving Physiotherapy services through MTM-field staff (n-320)	153	47.81%	(42.2-53.4)
Proportion receiving Dialysis services through MTM (n-43)	36	83.72%	(69.3-93.1)
Proportion receiving Dialysis services through MTM-field staff (n-43)	22	51.16%	(35.4-66.6)
Perception about MTM services			
Proportion who preferred home services under MTM (n-5184)	4726	91.17%	90.3-91.9
Proportion satisfied with MTM (n-5184)	4295	82.85%	81.8-83.8

STRATIFIED ANALYSIS OF KEY INDICATORS

Table 4. Proportion of people Aware of Makkalai Thedi Maruthuvam

Variable	Stratum	N (%)	95% CI	P value
Overall (N = 6856)		5405 (78.84%)	77.8-79.8%	
Gender (n- 6854)	Men (n- 2575)	2028 (78.76%)	77.1-80.3%	0.50
	Women (N= 4279)	3376 (78.90%)	77.6-80.1%	
Locality (n-6856)	Rural (N = 4155)	3461 (83.3%)	82.1-84.4%	0.001*
	Urban (N=2701)	1944 (71.9%)	70.2-73.6%	
Community groups (n-6856)	OC (N= 106)	62 (58.5%)	48.5-67.9%	0.001*
	BC (N = 2416)	1909 (79%)	77.3-80.6%	
	MBC (N = 1589)	1346 (84.7%)	82.8-86.4%	
	SC (N=825)	686 (83.15%)	80.4-85.6%	
	ST (N= 220)	192 (87.27%)	82.1-91.3%	
	Not willing to reveal (N= 1700)	1210 (71.2%)	68.9-73.3%	
Religion (n-6856)	Hindu (N = 6086)	4854 (79.8%)	78.7-80.7%	0.001*
	Muslim (N= 254)	166 (65.4%)	59.1-71.1%	
	Christian (N = 357)	298 (83.5%)	79.2-87.1%	
	Others (N = 7)	4 (57.1%)	18.4-90.1%	
	Not willing to reveal (N = 152)	83 (54.6%)	46.3-62.6%	
Economic class* (n-6856)	Class I (Upper Class) (N = 839)	631 (75.2%)	72.1-78.1%	0.04*
	Class II (Upper Middle Class) (N= 1336)	1044 (78.1%)	75.8-80.3%	
	Class III (Middle Class) (N= 2035)	1635 (80.3%)	78.5-82.0%	
	Class IV (Lower Middle Class) (N= 1457)	1155 (79.3%)	77.1-81.3%	
	Class V (Lower Class) (N= 1189)	940 (79.1%)	76.6-81.3%	
Education status (n-6856)	Illiterate (N = 1876)	1406 (74.9%)	72.9-76.8%	< 0.001*
	Primary Education (N = 1321)	1087 (82.3%)	80.1-84.3%	
	Middle school education (N = 1803)	1422 (78.9%)	76.9-80.7%	
	High school education (N = 977)	774 (79.2%)	76.5-81.7%	
	Diploma (N = 200)	167 (83.5%)	77.6-88.3%	
	Graduate (N = 679)	549 (80.9%)	77.6-83.7%	

- There is no gender difference in awareness.
- Rural people were more aware than urban people.
- There is a significant difference between the community with people belonging to ST, SC, MBC and BC more aware than those belonging to OC.
- Lower and middle economic class more aware compared to upper class.
- Illiterate less aware.

Table 5. Proportion ever visited by MTM field functionary.

Variable	Stratum	N (%)	95% CI	P value
Overall	N = 6856	4988 (72.7%)	71.6-73.8%	
Gender (n- 6854)	Men (N = 2575)	1821 (70.7%)	68.9-72.4%	0.01*
	Women (N = 4279)	3166 (74%)	72.6-75.3%	
Locality (n-6856)	Rural (N = 4155)	3437 (82.7%)	81.5-83.8%	< 0.001*
	Urban (N=2701)	1551 (57.4%)	55.5-59.3%	
community groups (n-6856)	OC (N= 106)	56 (52.8%)	42.8-62.6%	0.001*
	BC (N = 2416)	1689 (69.9%)	68.0-71.7%	
	MBC (N = 1589)	1293 (81.4%)	79.3-83.2%	
	SC (N=825)	671 (81.3%)	78.5-83.9%	
	ST (N= 220)	192 (87.3%)	82.1-91.3%	
	Not willing to reveal (N= 1700)	1087 (63.9%)	61.6-66.2%	
Religion (n-6856)	Hindu (N = 6086)	4568 (75.1%)	73.9-76.1%	< 0.001*
	Muslim (N= 254)	120 (47.2%)	40.9-53.5%	
	Christian (N = 357)	255 (71.4%)	66.4-76.0%	
	Others (N = 7)	2 (28.6%)	3.6-70.9%	
	Not willing to reveal (N = 152)	43 (28.3%)	21.2-36.1%	
Economic class* (n-6856)	Class I (Upper Class) (N = 839)	413 (49.2%)	45.7-52.6%	< 0.001*
	Class II (Upper Middle Class) (N= 1336)	939 (70.3%)	67.7-72.7%	
	Class III (Middle Class) (N= 2035)	1513 (74.3%)	72.3-76.2%	
	Class IV (Lower Middle Class) (N= 1457)	1139 (78.2%)	75.9-80.2%	
	Class V (Lower Class) (N= 1189)	984 (82.8%)	80.4-84.8%	
	Illiterate (N = 1876)	1474 (78.6%)	76.6-80.4%	
Education status (n-6856)	Primary Education (N = 1321)	1068 (80.8%)	78.6-82.9%	< 0.001*
	Middle School Education (N = 1803)	1291 (71.6%)	69.4-73.6%	
	High School Education (N = 977)	654 (66.9%)	63.8-69.8%	
	Diploma (N = 200)	128 (64%)	56.9-70.6%	
	Graduate (N = 679)	373 (54.9%)	51.1-58.7%	

- With regards to proportion visited by Women Health Volunteer
 - More women visited by a WHV.
 - Rural coverage more than urban
 - SC/ST/MBC covered more than BC/OC
 - Poor people covered more than upper class.
 - Illiterate and people with lower education status covered more than graduates.

Table 6. Proportion ever screened for Diabetes Mellitus

Variable	Stratum	N (%)	95% CI	P value
Overall (n-6856)		5448 (79.5%)	78.5-80.4%	
Gender	Men (N = 2575)	1988 (77.2 %)	75.5-78.8%	0.001*
	Women (N = 4279)	3459 (80.8%)	79.6-82.0%	
Locality (n-6856)	Rural (N = 4155)	3344 (80.5%)	79.2-81.6%	0.01*
	Urban (N=2701)	2104 (77.9%)	76.3-79.4%	
Community groups (n-6856)	OC (N= 106)	84 (79.2%)	70.2- 86.5%	>0.005
	BC (N = 2416)	1977 (81.8%)	80.2- 83.3%	
	MBC (N = 1589)	1293 (81.4%)	79.3- 83.2%	
	SC (N=825)	667 (80.9%)	77.9-83.4%	
	ST (N= 220)	179 (81.4%)	75.5-86.2%	
	Not willing to reveal (N= 1700)	1248 (73.4%)	71.2- 75.5%	
Religion (n-6856)	Hindu (N = 6086)	4833 (79.4%)	78.3- 80.4%	0.129
	Muslim (N= 254)	209 (82.3%)	77.0-86.7%	
	Christian (N = 357)	284 (79.6%)	74.9-83.6%	
	Others (N = 7)	3 (42.9%)	9.9- 81.5%	
	Not willing to reveal (N = 152)	119 (78.3%)	70.8- 84.5%	
Economic class* (n-6856)	Class I (Upper Class) (N = 839)	664 (79.1%)	76.2- 81.8%	<0.001*
	Class II (Upper Middle Class) (N= 1336)	1002 (75%)	72.5- 77.3%	
	Class III (Middle Class) (N= 2035)	1619 (79.6%)	77.7-81.2%	
	Class IV (Lower Middle Class) (N= 1457)	1204 (82.6%)	80.5-84.5%	
	Class V (Lower Class) (N= 1189)	959 (80.6%)	78.2-82.8%	
Education status (n-6856)	Illiterate (N = 1876)	1519 (80.9%)	79.1-82.7%	<0.001*
	Primary Education (N = 1321)	1091 (82.6%)	80.4-84.6%	
	Middle School Education (N = 1803)	1425 (79.0%)	77.0-80.8%	
	High School Education (N = 977)	756 (77.4%)	74.6-79.9%	
	Diploma (N = 200)	155 (77.5%)	71.0-83.2%	
	Graduate (N = 679)	502 (73.9%)	70.4-77.2%	

- With regards to screening for Diabetes
 - More women screened than men.
 - Rural coverage more than urban.
 - No significant difference between **Community** and religion group.
 - Screening coverage high among Poor people.
 - Illiterate/ people with lower education status covered more than graduates.

Table 7. Proportion screened for DM through MTM

Variable	Stratum	N (%)	95% CI	P value
Overall (n=5448)		3685 (67.6%)	66.3-68.8%	
Gender	Men (N=1988)	1293 (65.0%)	62.9-67.1%	0.006*
	Women (N=3459)	2391 (69.1%)	67.5-70.7%	
Locality (n=5448)	Rural (N=3344)	2574 (77.0%)	75.5-78.4%	0.000*
	Urban (N=2104)	1111 (52.8%)	50.6-54.9%	
Community groups (n=5448)	OC (N=84)	30 (35.7%)	25.5-46.9%	0.000*
	BC (N=1977)	1253 (63.4%)	61.2-65.5%	
	MBC (N=1293)	940 (72.7%)	70.2-75.1%	
	SC (N=667)	533 (79.9%)	76.7-82.9%	
	ST (N=179)	152 (84.9%)	78.8-89.8%	
	Not willing to reveal (N=1248)	777 (62.3%)	59.5-64.9%	
Religion (n=5448)	Hindu (N=4833)	3354 (69.4%)	68.1-70.7%	0.000*
	Muslim (N=209)	99 (47.4%)	40.4-54.4%	
	Christian (N=284)	185 (65.1%)	59.3-70.7%	
	Others (N=3)	1 (33.3%)	0.8-90.6%	
	Not willing to reveal (N=119)	46 (38.7%)	29.9-48.0%	
Economic class* (n=5448)	Class I (Upper Class) (N=664)	483 (72.7%)	69.2-76.1%	0.013*
	Class II (Upper Middle Class) (N=1002)	651 (65.0%)	61.9-67.9%	
	Class III (Middle Class) (N=1619)	1105 (68.3%)	65.9-70.5%	
	Class IV (Lower Middle Class) (N=1204)	814 (67.6%)	64.9-70.3%	
	Class V (Lower Class) (N=959)	632 (65.9%)	62.8-68.9%	
Education status (n=5448)	Illiterate (N=1519)	1198 (78.9%)	76.7-80.9%	0.000*
	Primary Education (N=1091)	802 (73.5%)	70.8-76.1%	
	Middle School Education (N=1425)	947 (66.5%)	63.9-68.9%	
	High School Education (N=756)	443 (58.6%)	54.9-62.1%	
	Diploma (N=155)	85 (54.8%)	46.6-62.8%	
	Graduate (N=502)	212 (42.2%)	37.8-46.7%	

- With regards to screening for Diabetes through MTM
 - More women screened than men.
 - Rural coverage more than urban.
 - SC/ST/MBC covered more than BC/OC
 - Screening coverage high among upper economic class through MTM
 - Illiterate/ people with lower education status covered more than graduates.

Table 8. Proportion screened for DM through MTM by field functionaries

Variable	Stratum	N (%)	95% CI	P value
Overall (n-3685)	-	2566 (69.6%)	68.1-71.1%	
Gender	Men (N=1293)	847 (65.5%)	62.8-68.1%	0.000*
	Women (N=2391)	1719 (71.9%)	70.0-73.6%	
Locality (n-3685)	Rural (N=2574)	1823 (70.8%)	69.0-72.5%	0.017*
	Urban (N=1111)	743 (66.9%)	64.0-69.6%	
Community groups (n-3685)	OC (N=30)	20 (66.7%)	47.2-82.7%	0.000*
	BC (N=1253)	898 (71.7%)	69.1-74.2%	
	MBC (N=940)	694 (73.8%)	70.9-76.6%	
	SC (N=533)	346 (64.9%)	60.7-68.9%	
	ST (N=152)	130 (85.5%)	78.9-90.7%	
	Not willing to reveal (N=777)	478 (61.5%)	57.9-64.9%	
Religion (n-3685)	Hindu (N=3354)	2362 (70.4%)	68.9-71.9%	0.000*
	Muslim (N=99)	52 (52.5%)	42.2-62.7%	
	Christian (N=185)	132 (71.4%)	64.3-77.8%	
	Others (N=1)	1 (100.0%)	2.5-100%	
	Not willing to reveal (N=46)	19 (41.3%)	27-56.8%	
Economic class* (n-3685)	Class I (Upper Class) (N=483)	361 (74.7%)	70.6-78.6%	0.001*
	Class II (Upper Middle Class) (N=651)	431 (66.2%)	62.4-69.8%	
	Class III (Middle Class) (N=1105)	735 (66.5%)	63.7-69.3%	
	Class IV (Lower Middle Class) (N=814)	578 (71.0%)	67.8-74.1%	
	Class V (Lower Class) (N=632)	461 (72.9%)	69.3-76.4%	
Education status (n-3685)	Illiterate (N=1198)	872 (72.8%)	70.2-75.3%	0.025*
	Primary Education (N=802)	546 (68.1%)	64.7-71.3%	
	Middle School Education (N=947)	640 (67.6%)	64.5-70.6%	
	High School Education (N=441)	299 (67.8%)	63.2-72.1%	
	Diploma (N=85)	53 (62.4%)	51.2-72.6%	
	Graduate (N=212)	156 (73.6%)	67.1-79.4%	

- With regards to screening for Diabetes through MTM field functionaries
 - More women screened than men.
 - Rural coverage more than urban.
 - ST covered more than other Communitygroups
 - Screening coverage high among upper class and lower class.

Table 9. Proportion ever diagnosed with Diabetes Mellitus

Variable	Stratum	N (%)	95% CI	P value
Overall (n-6856)		1419 (20.7%)	19.7-21.6%	
Gender	Men (N = 2575)	580 (22.5%)	20.9-24.2%	0.009*
	Women (N = 4279)	838 (19.6%)	18.4-20.8%	
Locality (n-6856)	Rural (N = 4155)	739 (17.8%)	16.6-8.9%	0.000*
	Urban (N=2701)	680 (25.2%)	23.5-26.9%	
Community groups (n-6856)	OC (N= 106)	35 (33.0%)	24.2-42.8%	0.000*
	BC (N = 2416)	573 (23.7%)	22.0-25.5%	
	MBC (N = 1589)	300 (18.9%)	16.9-20.9%	
	SC (N=825)	131 (15.9%)	13.5-18.6%	
	ST (N= 220)	21 (9.5%)	6.0-14.2%	
	Not willing to reveal (N= 1700)	359 (21.1%)	19.2-23.1%	
Religion (n-6856)	Hindu (N = 6086)	1215 (20.0%)	18.9-20.9%	0.000*
	Muslim (N= 254)	76 (29.9%)	24.4-35.9%	
	Christian (N = 357)	86 (24.1%)	19.7-28.9%	
	Others (N = 7)	2 (28.6%)	3.7-70.9%	
	Not willing to reveal (N = 152)	40 (26.3%)	19.5-34.1%	
Economic class* (n-6856)	Class I (Upper Class) (N = 839)	141 (16.8%)	14.3-19.5%	0.000*
	Class II (Upper Middle Class) (N= 1336)	234 (17.5%)	15.5-19.7%	
	Class III (Middle Class) (N= 2035)	437 (21.5%)	19.7-23.3%	
	Class IV (Lower Middle Class) (N= 1457)	323 (22.2%)	20.1-24.4%	
	Class V (Lower Class) (N= 1189)	284 (23.9%)	21.5-26.4%	
Education status (n-6856)	Illiterate (N = 1876)	382 (20.4%)	18.6-22.3%	0.003*
	Primary Education (N = 1321)	304 (23.0%)	20.8-25.4%	
	Middle School Education (N = 1803)	392 (21.7%)	19.9-23.7%	
	High School Education (N = 977)	186 (19.0%)	19.0-16.6%	
	Diploma (N = 200)	47 (23.5%)	17.8-30.0%	
	Graduate (N = 679)	108 (15.9%)	13.2-18.9%	

- **With regards to Diabetes proportion**

- More men than women.
- Urban more diabetic than rural.
- Higher among OC than others
- High among lower economic class.

Table 10. New diagnosis for DM in the last one year through MTM (among newly diagnosed)

Variable	Stratum	N (%)	95% CI	P value
Overall (n-236)	-	153 (64.8%)	58.3-70.9%	
Gender	Men (N = 87)	56 (64.4%)	53.3-74.3%	0.909
	Women (N = 149)	97(65.1%)	56.8-72.7%	
Locality (n-236)	Rural (N = 141)	106 (75.2%)	67.2-82.0%	0.000*
	Urban (N=95)	47 (49.5%)	39.0-59.9%	
Community groups (n-236)	OC (N= 6)	0 (0.0%)	0-45.9%	0.010*
	BC (N = 86)	51 (59.3%)	48.1-69.7%	
	MBC (N = 63)	43 (68.3%)	55.3-79.4%	
	SC (N=20)	14 (70.0%)	45.7-88.1%	
	ST (N= 5)	4 (80.0%)	28.3-99.4%	
	Not willing to reveal (N= 56)	41 (73.2%)	59.7-84.1%	
Religion (n-236)	Hindu (N = 207)	141 (68.1%)	61.3-74.4%	0.063
	Muslim (N= 12)	5 (41.7%)	15.1-72.3%	
	Christian (N = 13)	6 (46.2%)	19.2-74.8%	
	Others (N = 1)	0 (0.0%)	0-97.5%	
	Not willing to reveal (N = 3)	1 (33.3%)	0.8-90.5%	
Economic class* (n-236)	Class I (Upper Class) (N = 14)	10 (71.4%)	41.9-91.6%	0.320
	Class II (Upper Middle Class) (N= 41)	25 (61.0%)	44.5-75.8%	
	Class III (Middle Class) (N= 85)	60 (70.6%)	59.7-79.9%	
	Class IV (Lower Middle Class) (N= 49)	33 (67.3%)	52.4-80.0%	
	Class V (Lower Class) (N= 47)	25 (53.2%)	38.0-67.8%	
Education status (n-236)	Illiterate (N = 74)	57 (77.0%)	65.7-86.0%	0.001*
	Primary Education (N = 47)	34 (72.3%)	57.3-84.3%	
	Middle School Education (N = 67)	41 (61.2%)	48.5-72.8%	
	High School Education (N =30)	16 (53.3%)	34.3-71.6%	
	Diploma (N = 7)	3 (42.9%)	9.9-81.5%	
	Graduate (N = 11)	2 (18.2%)	2.2-51.7%	

- With regards new diagnosis of Diabetes mellitus in the last one year
 - More women were diagnosed than men.
 - More new cases in rural area than urban.
 - ST diagnosed more than other OC
 - More new cases diagnosed among upper class and middle class.
 - More proportion of illiterates have been diagnosed.

Table 11. Proportion on treatment for DM

Variable	Stratum	N (%)	95% CI	P value
Overall (n-1419)	-	1382 (97.4%)	96.4-98.1%	-
Gender	Men (N=580)	566 (97.6%)	95.9-98.7%	0.916
	Women (N=838)	815 (97.3%)	95.9-98.2%	
Locality (n-1419)	Rural (N=739)	712 (96.3%)	94.7-97.6%	0.01*
	Urban (N=680)	670 (98.5%)	97.3-99.3%	
Community groups (n-1419)	OC (N=35)	35 (100%)	90-100%	0.005*
	BC (N=573)	568 (99.1%)	98.0-99.7%	
	MBC (N=300)	285 (95.0%)	91.9-97.2%	
	SC (N=131)	126 (96.2%)	91.3-98.7%	
	ST (N=21)	21 (100%)	83.9-100%	
	Not willing to reveal (N=359)	347 (96.7%)	94.2-98.3%	
Religion (n-1419)	Hindu (N=1215)	1181 (97.2%)	96.1-98.1%	0.682
	Muslim (N=76)	76 (100%)	15.8-100%	
	Christian (N=86)	84 (97.7%)	86.8-99.9%	
	Others (N=2)	2 (100%)	15.8-100%	
	Not willing to reveal (N=40)	39 (97.5%)	86.8-99.9%	
Economic class* (n-1419)	Class I (Upper Class) (N=141)	139 (98.6%)	94.9-99.9%	0.555
	Class II (Upper Middle Class) (N=234)	226 (96.6%)	93.4-98.5%	
	Class III (Middle Class) (N=437)	426 (97.5%)	95.5-98.7%	
	Class IV (Lower Middle Class) (N=323)	312 (96.6%)	93.4-98.5%	
	Class V (Lower Class) (N=284)	279 (98.2%)	95.9-99.4%	
Education status (n-1419)	Illiterate (N=382)	375 (98.2%)	96.3-99.3%	0.586
	Primary Education (N=304)	295 (97.0%)	94.4-98.7%	
	Middle School Education (N=392)	379 (96.7%)	94.4-98.2%	
	High School Education (N=186)	182 (97.8%)	94.6-99.4%	
	Diploma (N=47)	47 (100%)	92.4-100%	
	Graduate (N=108)	104 (96.3%)	90.8-98.9%	

- With regards proportion on treatment for Diabetes mellitus
 - No gender differences.
 - Slightly higher treatment rate among urban.
 - ST and OC have higher proportion of treatment
 - Proportion of treatment is high among upper class and lower class.
 - Illiterates have a higher treatment proportion rate for Diabetes.

Table 12. Proportion on treatment for DM through MTM

Variable	Stratum	N (%)	95% CI	P value
Overall (n-1419)	-	774 (54.5%)	51.9-57.1%	-
Gender	Men (N=580)	306 (52.8%)	48.6-56.8%	0.332
	Women (N=838)	467 (55.7%)	52.2-59.1%	
Locality (n-1419)	Rural (N=739)	482 (65.2%)	61.6-68.6%	0.000*
	Urban (N=680)	292 (42.9%)	39.1-46.7%	
Community groups (n-1419)	OC (N=35)	11 (31.4%)	16.8-49.2%	0.000*
	BC (N=573)	301 (52.5%)	48.3-56.6%	
	MBC (N=300)	166 (55.3%)	49.5-61.0%	
	SC (N=131)	89 (67.9%)	59.2-75.8%	
	ST (N=21)	14 (66.7%)	43.0-85.4%	
	Not willing to reveal (N=359)	193 (53.7%)	48.4-59.0%	
Religion (n-1419)	Hindu (N=1215)	682 (56.1%)	53.2-58.9%	0.000*
	Muslim (N=76)	40 (52.6%)	40.8-64.2%	
	Christian (N=86)	43 (50.0%)	39.0-60.9%	
	Others (N=2)	0 (0%)	0-84.1%	
	Not willing to reveal (N=40)	9 (22.5%)	10.8-38.4%	
Economic class* (n-1419)	Class I (Upper Class) (N=141)	79 (56.0%)	47.4-64.3%	0.810
	Class II (Upper Middle Class) (N=234)	120 (51.3%)	44.6-57.8%	
	Class III (Middle Class) (N=437)	240 (54.9%)	50.1-59.6%	
	Class IV (Lower Middle Class) (N=323)	172 (53.3%)	47.6-58.8%	
	Class V (Lower Class) (N=284)	163 (57.4%)	51.4-63.2%	
Education status (n-1419)	Illiterate (N=382)	262 (68.6%)	63.6-73.2%	0.000*
	Primary Education (N=304)	184 (60.5%)	54.7-66.0%	
	Middle School Education (N=392)	212 (54.0%)	49.0-59.1%	
	High School Education (N=186)	74 (39.8%)	32.7-47.2%	
	Diploma (N=47)	20 (42.5%)	28.2-57.8%	
	Graduate (N=108)	22 (20.4%)	13.2-29.2%	

- With regards proportion on treatment for Diabetes mellitus through MTM
 - More women are covered through MTM than men.
 - More treatment in rural areas than urban through MTM
 - SC and ST have higher treatment coverage through MTM
 - Lower class have higher treatment coverage through MTM.
 - Illiterates and those with primary education have higher treatment coverage through MTM.

Table 13. Proportion of DM dispensed medicines through WHV for DM

Variable	Stratum	N (%)	95% CI	P value
Overall (n-1419)	-	511 (36.0%)	33.5-38.6%	-
Gender	Men (N=580)	190 (32.8%)	28.9-36.7%	0.228
	Women (N=838)	321 (38.3%)	35.0-41.7%	
Locality (n-1419)	Rural (N=739)	335 (45.3%)	41.7-49.0%	0.000*
	Urban (N=680)	176 (25.9%)	22.6-29.4%	
Community groups (n-1419)	OC (N=35)	7 (20.0%)	8.4-36.9%	0.001*
	BC (N=573)	202 (35.3%)	31.3-39.3%	
	MBC (N=300)	124 (41.3%)	35.7-47.1%	
	SC (N=131)	62 (47.3%)	38.5-56.2%	
	ST (N=21)	9 (42.9%)	21.8-65.9%	
	Not willing to reveal (N=359)	107 (29.8%)	25.1-34.8%	
Religion (n-1419)	Hindu (N=1215)	451 (37.1%)	34.4-39.9%	0.002*
	Muslim (N=76)	22 (28.9%)	19.1-40.4%	
	Christian (N=86)	36 (41.9%)	31.3-52.9%	
	Others (N=2)	0 (0.0%)	-	
	Not willing to reveal (N=40)	2 (5.0%)	0.6-1.7%	
Economic class* (n-1419)	Class I (Upper Class) (N=141)	53 (37.6%)	29.6-46.1%	0.028*
	Class II (Upper Middle Class) (N=234)	71 (30.3%)	24.5-36.6%	
	Class III (Middle Class) (N=437)	149 (34.1%)	29.7-38.8%	
	Class IV (Lower Middle Class) (N=323)	114 (35.3%)	30.1-40.7%	
	Class V (Lower Class) (N=284)	124 (43.7%)	37.8-49.7%	
Education status (n-1419)	Illiterate (N=382)	179 (46.9%)	41.7-52.0%	0.000*
	Primary Education (N=304)	122 (40.1%)	34.6-45.9%	
	Middle School Education (N=392)	131 (33.4%)	28.7-38.3%	
	High School Education (N=186)	54 (29.0%)	22.6-36.1%	
	Diploma (N=47)	11 (23.4%)	12.3-38.0%	
	Graduate (N=108)	14 (13.0%)	7.3-20.8%	

- With regards proportion of Diabetes mellitus dispensed medicine through WHV,
 - More women than men.
 - More dispensing of medicines in rural areas than urban by WHV
 - SC and ST have higher dispensing of medicines by WHV
 - Lower class have higher dispensing of medicines by WHV
 - Illiterates and those with primary education have dispensing of medicines by WHV.

Table 14. Proportion of DM with good glycemic control

Variable	Stratum	N (%)	95% CI	P value
Overall (n-1419)	-	139 (9.8%)	8.3-11.4%	
Gender	Men (N=580)	55 (9.5%)	7.2-12.1%	0.344
	Women (N=838)	84 (10.0%)	8.0-12.2%	
Locality (n-1419)	Rural (N=739)	63 (8.5%)	6.6-10.7%	0.244
	Urban (N=680)	76 (11.2%)	8.9-13.7%	
Community groups (n-1419)	OC (N=35)	5 (14.3%)	4.8-30.2%	0.018*
	BC (N=573)	53 (9.2%)	7.0-11.9%	
	MBC (N=300)	30 (10.0%)	6.8-13.9%	
	SC (N=131)	11 (8.4%)	4.2-14.5%	
	ST (N=21)	1 (4.8%)	0.1-23.8%	
	Not willing to reveal (N=359)	39 (10.9%)	7.8-14.5%	
Religion (n-1419)	Hindu (N=1215)	114 (9.4%)	7.8-11.1%	0.000*
	Muslim (N=76)	5 (6.6%)	2.1-14.6%	
	Christian (N=86)	9 (10.5%)	4.9-18.9%	
	Others (N=2)	1 (50.0%)	12.6-98.7%	
	Not willing to reveal (N=40)	10 (25.0%)	12.6-41.2%	
Economic class* (n-1419)	Class I (Upper Class) (N=141)	18 (12.8%)	7.7-19.4%	0.310
	Class II (Upper Middle Class) (N=234)	24 (10.3%)	6.6-14.8%	
	Class III (Middle Class) (N=437)	41 (9.4%)	6.8-12.5%	
	Class IV (Lower Middle Class) (N=323)	26 (8.0%)	5.3-11.5%	
	Class V (Lower Class) (N=284)	30 (10.6%)	7.2-14.7%	
Education status (n-1419)	Illiterate (N=382)	43 (11.3%)	8.2-14.8%	0.001*
	Primary Education (N=304)	22 (7.2%)	4.5-10.7%	
	Middle School Education (N=392)	26 (6.6%)	4.3-9.5%	
	High School Education (N=186)	30 (16.1%)	11.1-22.2%	
	Diploma (N=47)	5 (10.6%)	3.5-23.1%	
	Graduate (N=108)	13 (12.0%)	6.5-19.7%	

- With regards proportion of Diabetes mellitus with good glycemic control,
 - Women have better control than men.
 - Urban area have better control rates than rural
 - MBC have better control rates
 - Upper class have better glycemic control
 - Graduates have better glycemic control.

Table 15. Proportion of DM patients who shifted from private to public hospitals among those who reported shifting after MTM

Variable	Stratum	N (%)	95% CI	P value
Overall (n-331)	-	97 (29.3%)	24.4-34.5%	
Gender	Men (N=126)	38 (30.2%)	22.3-38.9%	0.789
	Women (N=205)	59 (28.8%)	22.6-35.5%	
Locality (n-331)	Rural (N=175)	55 (31.4%)	24.6-38.8%	0.368
	Urban (N=156)	42 (26.9%)	20.1-34.6%	
Community groups (n-331)	OC (N=10)	2 (20.0%)	2.5-55.6%	0.919
	BC (N=131)	48 (36.6%)	28.4-45.5%	
	MBC (N=73)	20 (27.4%)	17.6-39.0%	
	SC (N=43)	8 (18.6%)	8.3-33.4%	
	ST (N=3)	2 (66.7%)	9.4-99.1%	
	Not willing to reveal (N=71)	17 (23.9%)	14.6-35.5%	
Religion (n-331)	Hindu (N=278)	86 (30.9%)	25.5-36.7%	0.028*
	Muslim (N=20)	6 (30.0%)	11.8-54.2%	
	Christian (N=28)	4 (14.3%)	4.0-32.6%	
	Others (N=0)	0 (0.0%)	0	
	Not willing to reveal (N=5)	1 (20.0%)	5.1-71.6%	
Economic class* (n-331)	Class I (Upper Class) (N=38)	12 (31.6%)	17.5-48.6%	0.460
	Class II (Upper Middle Class) (N=52)	13 (25.0%)	14.0-38.9%	
	Class III (Middle Class) (N=107)	35 (32.7%)	23.9-42.4%	
	Class IV (Lower Middle Class) (N=92)	26 (28.3%)	19.3-38.6%	
	Class V (Lower Class) (N=42)	11 (26.2%)	13.8-42.0%	
Education status (n-331)	Illiterate (N=85)	23 (27.1%)	17.9-37.7%	0.162
	Primary Education (N=74)	24 (32.4%)	22.0-44.3%	
	Middle School Education (N=99)	32 (32.3%)	23.2-42.4%	
	High School Education (N=40)	9 (22.5%)	10.8-38.4%	
	Diploma (N=5)	3 (60.0%)	14.6-94.7%	
	Graduate (N=28)	6 (21.4%)	8.3-40.9%	

- With regards proportion of Diabetes mellitus patients shifted from private to public hospitals,
 - More men than women.
 - More shift in rural areas
 - More shift among SC and BC
 - More shift among Middle class and upper class
 - More shift among Diploma holders and primary education.

Table 16. Proportion ever screened for HTN

Variable	Stratum	N (%)	95% CI	P value
Overall (n-6856)	-	5570 (81.3%)	80.3-82.1%	
Gender (n-6856)	Men (N=2574)	2036 (79.1%)	77.5-80.7%	0.000*
	Women (N=4279)	3534 (82.6%)	81.4-83.7%	
Locality (n-6856)	Rural (N=4155)	3427 (82.5%)	81.3-83.6%	0.001*
	Urban (N=2700)	2143 (79.4%)	77.8-80.9%	
Community groups (n-6856)	OC (N=106)	83 (78.3%)	69.2-85.7%	0.000*
	BC (N=2416)	2021 (83.7%)	82.1-85.1%	
	MBC(N=1589)	1332 (83.8%)	81.9-85.6%	
	SC (N=825)	671 (81.3%)	78.5-83.9%	
	ST (N=220)	187 (85%)	77.6-87.8%	
	Not willing to reveal (N=1699)	1276 (75.1%)	73-77.1%	
Religion (n-6856)	Hindu (N=6085)	4930 (81%)	80.0-82%	0.001*
	Muslim (N=254)	225 (88.6%)	84-92.2%	
	Christian (N=357)	296 (82.9%)	78.6-86.7%	
	Others (N=7)	3 (42.9%)	9.9-81.6%	
	Not willing to reveal (N=152)	116 (76.3%)	68.75-82.8%	
Economic class* (n-6856)	Class I (Upper Class) (N=838)	654(78%)	75-80.8%	0.149
	Class II (Upper Middle Class) (N=1336)	1095(82%)	79.8-84%	
	Class III (Middle Class) (N=2035)	1655(81.3%%)	79.6-83%	
	Class IV (Lower Middle Class)(N=1457)	1193(81.9%%)	79.8-83.8%	
	Class V (Lower Class) (N=1189)	973(81.8%%)	79.5-84%	
Education status (n-6856)	Primary Education (N=1321)	1113(84.3%)	82.2-86.2%	0.007*
	Middle School Education (N=1803)	1454(80.6%)	78.7-82.4%	
	High School Education (N=976)	772(79.1%)	76.4-81.6%	
	Diploma (N=200)	160(80%)	73.8-85.3%	
	Graduate (N=679)	532(78.4%)	75-81.4%	
	Illiterate (N=1876)	1539(82%)	80.2-83.7%	

- With regards to screening for Hypertension
 - More women screened than men.
 - Rural coverage more than urban.
 - ST coverage is higher
 - Upper middle class and lower middle class have higher coverage
 - Those with primary education and illiterates have higher coverage.

Table 17. Proportion screened for HTN through MTM in the last 1 year

Variable	Stratum	N (%)	95% CI	P value
Overall (n=5225)		3847 (73.6%)	72.4-74.8%	
Gender (n=5225)	Men (N=1910)	1359 (71.2%)	69.0-73.1%	0.002*
	Women (N=3315)	2488 (75.1%)	73.5-76.5%	
Locality (n=5225)	Rural (N=3247)	2691 (82.9%)	81.5-84.1%	0.000*
	Urban (N=1878)	1156 (61.5%)	59.3-63.7%	
Community groups (n=5225)	OC (N=77)	31 (40.3%)	29.2-52.0%	0.000*
	BC (N=1919)	1288 (67.1%)	64.9-69.2%	
	MBC (N=1257)	982 (78.1%)	75.7-80.3%	
	SC (N=638)	558 (87.5%)	84.6-89.9%	
	ST (N=175)	162 (92.6%)	87.6-95.9%	
	Not willing to reveal (N=1159)	826 (71.3%)	68.5-73.8%	
Religion (n=5225)	Hindu (N=4626)	3499 (75.6%)	74.3-76.8%	0.000*
	Muslim (N=206)	106 (51.5%)	44.4-58.4%	
	Christian (N=279)	193 (69.2%)	63.3-74.5%	
	Others (N=3)	1 (33.3%)	0.8-90.5%	
	Not willing to reveal (N=111)	48 (43.2%)	33.8-52.9%	
Economic class* (n=5225)	Class I (Upper Class) (N=600)	269 (44.8%)	40.8-48.9%	0.000*
	Class II (Upper Middle Class)(N=1032)	709 (68.7%)	65.7-71.5%	
	Class III (Middle Class)(N=1552)	1198 (77.2%)	75.0-79.2%	
	Class IV (Lower Middle Class)(N=1125)	904 (80.4%)	77.9-82.6%	
	Class V (Lower Class)(N=916)	767 (83.7%)	81.1-86.0%	
Education status (n=5225)	Illiterate (N=1477)	1243 (84.2%)	82.1-85.9%	0.000*
	Primary Education (N=1031)	836 (81.1%)	78.5-83.4%	
	Middle School Education (N=1352)	971 (71.8%)	69.3-74.2%	
	High School Education (N=720)	475 (66%)	62.3-69.4%	
	Diploma (N=154)	88 (57.1%)	48.9-65.0%	
	Graduate (N=491)	234 (47.4%)	43.1-52.1%	

- With regards to screening for Hypertension through MTM
 - More women screened than men.
 - Rural coverage more than urban.
 - Coverage is more than sixty five percent among all Communities
 - Screening coverage high among lower economic class through MTM
 - Illiterate/ people with lower education status covered more than graduates.

Table 18. Proportion screened for HTN through MTM by field functionaries

Variable	Stratum	N (%)	95% CI	P value
Overall (n=3847)		2732 (71%)	69.5-72.4%	
Gender (n=3847)	Men (N=2488)	1804 (72.5%)	70.7-74.2%	0.003*
	Women (N=1359)	928 (68.3%)	65.7-70.7%	
Locality (n=3847)	Rural (N=2691)	1965 (73%)	71.3-74.6%	0.000*
	Urban (N=1156)	767 (66.3%)	63.5-69.0%	
Community groups (n=3847)	OC (N=31)	20 (64.5%)	45.3-80.7%	0.000*
	BC (N=1288)	931 (72.3%)	69.7-74.7%	
	MBC (N=982)	738 (75.2%)	72.3-77.8%	
	SC (N=558)	361 (64.7%)	60.5-68.6%	
	ST (N=162)	148 (91.4%)	85.9-95.1%	
	Not willing to reveal (N=826)	534 (64.6%)	61.3-67.9%	
Religion (n=3847)	Hindu (N=3499)	2502 (71.5%)	69.9-73.0%	0.000*
	Muslim (N=106)	57 (53.8%)	43.8-63.5%	
	Christian (N=193)	149 (77.2%)	70.6-82.9%	
	Others (N=1)	1 (100%)	2.5-100%	
	Not willing to reveal (N=48)	23 (47.9%)	33.2-62.8%	
Economic class* (n=3847)	Class I (Upper Class) (N=269)	181 (67.3%)	61.3-72.8%	0.000*
	Class II (Upper Middle Class) (N=709)	485 (68.4%)	64.8-71.8%	
	Class III (Middle Class) (N=1198)	801 (66.9%)	64.1-69.5%	
	Class IV (Lower Middle Class) (N=904)	659 (72.9%)	69.8-75.7%	
	Class V (Lower Class) (N=767)	606 (79%)	75.9-81.8%	
Education status (n=3847)	Illiterate (N=1243)	934 (75.1%)	72.6-77.5%	0.003*
	Primary Education (N=836)	578 (69.1%)	65.8-72.2%	
	Middle School Education (N=971)	669 (68.9%)	65.8-71.8%	
	High School Education (N=475)	321 (67.6%)	63.1-71.7%	
	Diploma (N=88)	59 (67%)	56.2-76.7%	
	Graduate (N=234)	171 (73.1%)	66.9-78.6%	

- With regards to screening for Hypertension through MTM field functionaries
 - More men screened than women.
 - Rural coverage more than urban.
 - ST covered more than OC
 - Screening coverage high among lower class and lower middle class
 - Screening coverage high among illiterates.

Table 19. Proportion ever diagnosed with HTN

Variable	Stratum	N (%)	95% CI	P value
Overall (n=6856)		1511 (22%)	21.1-23%	
Gender	Men (N=2574)	557(21.6%)	20.1-23.3%	0.613
	Women (N=4279)	954 (22.3%)	20.1-22.5%	
Locality (n=6856)	Rural (N=4155)	833 (20%)	18.8-21.3%	0.000*
	Urban (N=2701)	678 (25.1%)	23.5-26.8%	
Community groups (n=6856)	OC (N=106)	30 (28.3%)	20-37.9%	0.000*
	BC (N=2416)	621 (25.7%)	24-27.5%	
	MBC(N=1589)	319 (20.1%)	18.1-22.1%	
	SC (N=825)	151 (18.3%)	15.7-21.1%	
	ST (N=220)	47 (21.4%)	16.1-27.4%	
	Not willing to reveal (N=1700)	343 (20.2%)	18.3-22.2%	
Religion (n=6856)	Hindu (N=6086)	1312 (21.6%)	20.5-22.6%	0.000*
	Muslim (N=254)	88 (34.6%)	28.8-40.8%	
	Christian (N=357)	71 (19.9%)	15.9-24.4%	
	Others (N=7)	2 (28.6%)	3.7-71%	
	Not willing to reveal (N=152)	38 (25%)	18.3-32.7%	
Economic class* (n=6856)	Class I (Upper Class) (N=839)	213 (25.4%)	22.7-28.5%	0.000*
	Class II (Upper Middle Class) (N=1336)	283 (21.2%)	19-23.5%	
	Class III (Middle Class) (N=2035)	395 (19.4%)	17.7-21.2%	
	Class IV (Lower Middle Class)(N=1457)	313 (21.5%)	19.4-23.7%	
	Class V (Lower Class) (N=1189)	307 (25.8%)	23.3-28.4%	
Education status (n=6856)	Primary Education (N=1321)	313 (23.7%)	21.4-26.1%	0.000*
	Middle School Education (N=1803)	360 (20%)	18.1-22%	
	High School Education (N=976)	182 (18.6%)	16.2-21.2%	
	Diploma (N=200)	36 (18%)	12.9-24%	
	Graduate (N=679)	93 (13.7%)	11.2-16.5%	
	Illiterate (N=1876)	527 (28.1%)	26.1-30.2%	

- **With regards to Hypertensives proportion**

- More women than men.
- Urban more hypertensive than rural.
- Higher among OC than others
- High among lower economic class.
- Higher proportion among illiterates.

Table 20. New diagnosis for HTN in the last one year through MTM (among newly diagnosed)

Variable	Stratum	N (%)	95% CI	P value
Overall (n-289)		218 (75.4%)	70-80.3%	
Gender (n-289)	Men (N=105)	76 (72.4%)	62.8-80.7%	0.363
	Women (N=184)	142 (77.2%)	70.4-83%	
Locality (n-289)	Rural (N=179)	149 (89.2%)	77-88.4%	0.000*
	Urban (N=110)	69 (62.7%)	53-71.8%	
Community groups (n-289)	OC (N=3)	1 (33.3%)	0.8-90.6%	0.038*
	BC (N=108)	80 (74.1%)	64.7-82%	
	MBC (N=73)	58 (73.5%)	68.4-88%	
	SC (N=34)	29 (85.3%)	68.9-95%	
	ST (N=15)	14 (93.3%)	68-99.8%	
	Not willing to reveal (N=56)	36 (64.3%)	50.4-76.6%	
Religion (n-289)	Hindu (N=247)	188 (76.1%)	70.3-81.3%	0.561
	Muslim (N=23)	16 (69.6%)	47.1-86.8%	
	Christian (N=15)	12 (80%)	51.9-95.7%	
	Others (N=0)	0 (%)	0%	
	Not willing to reveal (N=4)	2 (50%)	6.8-93.2%	
Economic class* (n-289)	Class I (Upper Class) (N=21)	10 (47.6%)	25.7-70.2%	0.003*
	Class II (Upper Middle Class) (N=54)	36 (66.7%)	52.5-78.9%	
	Class III (Middle Class) (N=83)	62 (74.7%)	64-83.6%	
	Class IV (Lower Middle Class) (N=64)	54 (84.4%)	73.1-92.2%	
	Class V (Lower Class) (N=67)	56 (83.6%)	72.5-91.5%	
Education status (n-289)	Illiterate (N=108)	91 (84.3%)	76-90.5%	0.003*
	Primary Education (N=60)	47 (78.3%)	65.8-87.9%	
	Middle School Education (N=68)	49 (72.1%)	59.8-82.3%	
	High School Education (N=32)	21 (65.6%)	46.8-89.4%	
	Diploma (N=5)	1 (20%)	0.5-71.6%	
	Graduate (N=16)	9 (56.3%)	29.9-80.2%	

- With regards new diagnosis of hypertension in the last one year
 - More women were diagnosed than men.
 - More new cases in rural area than urban.
 - ST diagnosed more than other OC
 - More new cases diagnosed among lower class and lower middle class.
 - More proportion of illiterates have been diagnosed.

Table 21. Proportion on treatment for HTN

Variable	Stratum	N (%)	95% CI	P value
Overall (n-1511)		1454 (96.2%)	95.1-97.1%	
Gender (n-1511)	Men (N=557)	533 (95.7%)	93.7-97.2%	0.241
	Women (N=954)	921 (96.5%)	95.2-97.7%	
Locality (n-1511)	Rural (N=833)	798 (95.8%)	94.2-97.1%	0.332
	Urban (N=678)	656 (96.8%)	95.1-98%	
Community groups (n-1511)	OC (N=30)	29 (96.9%)	82.8-99.9%	0.949
	BC (N=621)	599 (96.5%)	94.7-97.8%	
	MBC (N=319)	306 (95.9%)	93.1-97.8%	
	SC (N=151)	145 (96%)	91.5-98.5%	
	ST (N=47)	44 (93.6%)	82.5-98.7%	
	Not willing to reveal (N=343)	331 (96.5%)	94-98.2%	
Religion (n-1511)	Hindu (N=1312)	1262 (96.2%)	95-97.2%	0.969
	Muslim (N=88)	84 (95.5%)	88.8-98.7%	
	Christian (N=71)	69 (97.2%)	90.2-99.7%	
	Others (N=2)	2 (100%)	15.8-100%	
	Not willing to reveal (N=37)	97.4 (%)	86.2-99.9%	
Economic class* (n-1511)	Class I (Upper Class) (N=213)	208 (97.7%)	94.6-99.2%	0.629
	Class II (Upper Middle Class) (N=283)	274 (96.8%)	94-98.5%	
	Class III (Middle Class) (N=395)	377 (95.4%)	92.9-97.3%	
	Class IV (Lower Middle Class) (N=313)	299 (95.5%)	92.6-97.5%	
	Class V (Lower Class) (N=307)	296 (96.4%)	93.7-98.2%	
Education status (n-1511)	Illiterate (N=527)	513 (97.3%)	95.6-98.5%	0.142
	Primary Education (N=313)	303 (96.8%)	94.2-98.4%	
	Middle School Education (N=360)	347 (96.4%)	93.9-98.1%	
	High School Education (N=182)	170 (93.4%)	88.8-96.4%	
	Diploma (N=36)	34 (96.4%)	81.3-99.3%	
	Graduate (N=93)	87 (93.5%)	86.5-97.6%	

- With regards proportion on treatment for Hypertension
 - More among women than men
 - Slightly higher treatment rate among urban.
 - OC and BC have higher proportion of treatment
 - Proportion of treatment is high among upper class and upper middle class.
 - Illiterates have a higher treatment proportion rate for Hypertension.

Table 22. Proportion on treatment for HTN through MTM

Variable	Stratum	N (%)	95% CI	P value
Overall (n-1511)		947 (62.67%)	60.2-65.1%	
Gender (n-1511)	Men (N=557)	344 (61.8%)	57.6-65.8%	0.574
	Women (N=954)	603 (63.2%)	60.1-66.3%	
Locality (n-1511)	Rural (N=833)	621 (74.5%)	71.5-77.5%	0.000*
	Urban (N=678)	326 (48.1%)	44.3-52%	
Community groups (n-1511)	OC (N=30)	9 (30%)	14.7-49.4%	0.000*
	BC (N=621)	357 (57.5%)	53.5-61.4%	
	MBC (N=319)	210 (65.8%)	60.3-71%	
	SC (N=151)	120 (79.5%)	72.1-85.6%	
	ST (N=47)	37 (78.7%)	64.3-89.3%	
	Not willing to reveal (N=343)	214 (61.4%)	57-67.5%	
Religion (n-1511)	Hindu (N=1312)	850 (64.8%)	62.1-67.3%	0.000*
	Muslim (N=88)	41 (46.6%)	35-57.5%	
	Christian (N=71)	38 (53.5%)	41.3-65.4%	
	Others (N=2)	1 (50%)	0.1-98.7%	
	Not willing to reveal (N=37)	17 (44.7%)	29.5-63%	
Economic class* (n-1511)	Class I (Upper Class) (N=213)	77 (36.2%)	29.7-43%	0.000*
	Class II (Upper Middle Class) (N=283)	167 (59%)	53-64.8%	
	Class III (Middle Class) (N=395)	251 (63.5%)	58.6-68.3%	
	Class IV (Lower Middle Class) (N=313)	213 (68.1%)	62.6-73.2%	
	Class V (Lower Class) (N=307)	239 (77.9%)	72.8-82.4%	
Education status (n-1511)	Illiterate (N=527)	405 (76.9%)	73-80.4%	0.000*
	Primary Education (N=313)	217 (69.3%)	63.9-74.4%	
	Middle School Education (N=360)	206 (57.2%)	51.9-62.4%	
	High School Education (N=182)	87 (47.8)	40.4-55.3%	
	Diploma (N=36)	13 (36.1%)	20.8-53.8%	
	Graduate (N=93)	19 20.4(%)	12.8-30.1%	

- With regards proportion on treatment for Hypertension through MTM
 - More women are covered through MTM than men.
 - More treatment in rural areas than urban through MTM
 - ST and SC have higher treatment coverage through MTM
 - Lower class have higher treatment coverage through MTM.
 - Illiterates and those with primary education have higher treatment coverage through MTM.

Table 23. Proportion of HTN dispensed medicines through WHV

Variable	Stratum	N (%)	95% CI	P value
Overall (n-1511)		684 (45.3%)	42.7-47.8%	
Gender (n-1511)	Men (N=557)	236 (42.4%)	38.2-46.6%	0.084
	Women (N=954)	448 (47%)	43.8-50.2%	
Locality (n-1511)	Rural (N=833)	480 (57.6%)	54.2-61%	0.000*
	Urban (N=678)	204 (30.1%)	26.7-33.7%	
Community groups (n-1511)	OC (N=30)	7 (23.3%)	10-42.3%	0.000*
	BC (N=621)	268 (43.2%)	39.2-47.2%	
	MBC (N=319)	167 (52.4%)	46.7-57.9%	
	SC (N=151)	86 (57%)	48.7-65%	
	ST (N=47)	34 (72.3%)	57.4-84.4%	
	Not willing to reveal (N=343)	122 (35.6%)	30.5-40.9%	
Religion (n-1511)	Hindu (N=1312)	625 (47.6%)	44.9-50.4%	0.000*
	Muslim (N=88)	28 (31.8%)	22.3-42.6%	
	Christian (N=71)	26 (36.6%)	25.5-48.9%	
	Others (N=2)	0(%)	0%	
	Not willing to reveal (N=37)	5 (13.2%)	4.5-28.8%	
Economic class* (n-1511)	Class I (Upper Class) (N=213)	52 (24.4%)	18.8-30.8%	0.000*
	Class II (Upper Middle Class) (N=283)	118 (41.7%)	35.9-47.7%	
	Class III (Middle Class) (N=395)	168 (42.5%)	37.6-47.6%	
	Class IV (Lower Middle Class) (N=313)	158 (50.5%)	44.8-56.2%	
	Class V (Lower Class) (N=307)	188 (61.2%)	55.5-66.7%	
Education status (n-1511)	Illiterate (N=527)	317 (60.2%)	55.8-64.4%	0.000*
	Primary Education (N=313)	157 (50.2%)	44.5-55.8%	
	Middle School Education (N=360)	130 (36.1%)	31.1-41.3%	
	High School Education (N=182)	58 (31.9%)	25.1-39.2%	
	Diploma (N=36)	8 (22.2%)	10.1-39.2%	
	Graduate (N=93)	14 (15.1%)	8.5-24%	

- With regards proportion of Hypertension dispensed medicine through WHV,
 - More women than men.
 - More dispensing of medicines in rural areas than urban by WHV
 - ST have higher dispensing of medicines by WHV
 - Lower class have higher dispensing of medicines by WHV
 - Illiterates and those with primary education have higher dispensing of medicines by WHV.

Table 24. Proportion of HTN with good hypertension control

Variable	Stratum	N (%)	95% CI	P value
Overall (n-1511)		535 (35.4%)	33-37.9%	
Gender (n-1511)	Men (N=557)	189 (33.9%)	30-38%	0.360
	Women (N=954)	346 (36.3%)	33.2-39.4%	
Locality (n-1511)	Rural (N=833)	296 (35.5%)	32.3-38.9%	0.909
	Urban (N=678)	239 (35.3%)	31.7-39%	
Community groups (n-1511)	OC (N=30)	14 (46.7%)	28.3-65.7%	0.297
	BC (N=621)	219 (35.3%)	31.5-39.2%	
	MBC (N=319)	126 (39.5%)	34.1-45.1%	
	SC (N=151)	51 (33.8%)	26.3-41.9%	
	ST (N=47)	15 (31.9%)	19-47.1%	
	Not willing to reveal (N=343)	110 (32.1%)	47.2-37.3%	
Religion (n-1511)	Hindu (N=1312)	475 (36.2%)	33.6-38.9%	0.173
	Muslim (N=88)	24 (27.3%)	18.3-37.8%	
	Christian (N=71)	20 (28.2%)	18.1-40.1%	
	Others (N=2)	0 (%)	0%	
	Not willing to reveal (N=37)	16 (42.1%)	27.1-60.5%	
Economic class* (n-1511)	Class I (Upper Class) (N=213)	85 (39.9%)	33.3-46.8%	0.189
	Class II (Upper Middle Class) (N=283)	87 (30.7%)	25.4-36.5%	
	Class III (Middle Class) (N=395)	139 (35.2%)	30.5-40.1%	
	Class IV (Lower Middle Class) (N=313)	106 (33.9%)	28.6-39.4%	
	Class V (Lower Class) (N=307)	118 (38.4%)	33-44.1%	
Education status (n-1511)	Illiterate (N=527)	204 (38.7%)	34.5-43%	0.167
	Primary Education (N=313)	108 (34.5%)	29.3-40%	
	Middle School Education (N=360)	120 (33.3%)	28.5-38.5%	
	High School Education (N=182)	58 (31.9%)	25.2-39.2%	
	Diploma (N=36)	8 (22.2%)	10.1-39.2%	
	Graduate (N=93)	37 (39.8%)	29.8-50.5%	

- With regards proportion of Hypertension with good control,
 - Women have better control than men.
 - No differences in locality
 - OC have better control rates
 - Upper class have better control rates
 - Graduates have better control rates

Table 25. Proportion of HTN patients who shifted from private to public hospitals

Variable	Stratum	N (%)	95% CI	p Value
Overall (n-292)		77(26.4%)	21.4-32%	
Gender (n-292)	Men (N=126)	33 (26.2%)	18.8-34.8%	0.952
	Women (N=166)	44 (26.5%)	20-33.9%	
Locality (n-292)	Rural (N=220)	59 (26.8%)	21.1-33.2%	0.761
	Urban (N=72)	18 (25%)	15.5-36.6%	
Community groups (n-292)	OC (N=1)	0 (%)	0%	0.585
	BC (N=103)	25 (24.3%)	16.4-33.7%	
	MBC (N=78)	21 (26.9%)	17.5-38.2%	
	SC (N=29)	10 (34.5%)	17.9-54.3%	
	ST (N=21)	8 (38.1%)	18.1-61.6%	
	Not willing to reveal (N=60)	13 (21.7%)	12.1-34.2%	
Religion (n-292)	Hindu (N=275)	75 (27.3%)	22.1-32.9%	0.667
	Muslim (N=5)	1(20%)	0.5-71.6%	
	Christian (N=9)	1(11.1%)	0.3-48.3%	
	Others (N=1)	0 (%)	0%	
	Not willing to reveal (N=2)	0 (%)	0%	
Economic class* (n-292)	Class I (Upper Class) (N=31)	4(12.9%)	3.6-29.8%	0.403
	Class II (Upper Middle Class) (N=56)	14(25%)	14.4-38.4%	
	Class III (Middle Class) (N=84)	23(27.4%)	78.9-99.9%	
	Class IV (Lower Middle Class) (N=66)	21(31.8%)	20.9-44.4%	
	Class V (Lower Class) (N=55)	15(27.3%)	16.1-41%	
Education status (n-292)	Illiterate (N=75)	28 (37.3%)	26.4-49.3%	0.227
	Primary Education (N=54)	14 (25.9%)	15-39.7%	
	Middle School Education (N=81)	17 (21%)	12.7-31.5%	
	High School Education (N=46)	11 (23.9%)	12.6-38.8%	
	Diploma (N=6)	1 (16.7%)	0.4-64.1%	
	Graduate (N=30)	6 (20%)	7.7-38.6%	

- With regards proportion of Hypertensive patients shifted from private to public hospitals,
 - No differences in gender.
 - More shift in rural areas
 - More shift among ST and SC
 - More shift among Lower Middle class and middle class
 - More shift among Illiterates.

Table 26. Proportion advised to get Oral Cancer Screening

Variable	Stratum	N (%)	95% CI	P value
Overall	N = 6856	1168 (17.0%)	16.1-17.9%	-
Gender	Men (N= 2575)	416 (16.2%)	14.7-17.6%	0.26
	Women (N= 4279)	752 (17.6%)	16.4-18.7%	
Locality (n-6856)	Rural (N = 4155)	709 (17.1%)	15.9-18.2%	0.94
	Urban (N=2701)	459 (17%)	15.6-18.4%	
Community groups (n-6856)	OC (N= 106)	17 (16%)	9.6-24.4%	0.02*
	BC (N = 2416)	421 (17.4%)	15.9-19.0%	
	MBC (N = 1589)	268 (16.9%)	15.0-18.8%	
	SC (N=825)	160 (19.4%)	16.7-22.2%	
	ST (N= 220)	21 (9.5%)	6.0-14.2%	
	Not willing to reveal (N= 1700)	281 (16.5%)	14.7-18.3%	
Religion (n-6856)	Hindu (N = 6086)	1060 (17.4%)	16.4-18.3%	0.002*
	Muslim (N= 254)	32 (12.6%)	8.7-17.3%	
	Christian (N = 357)	40 (11.2%)	8.1-14.9%	
	Others (N = 7)	1 (14.3%)	0.3-57.8%	
	Not willing to reveal (N = 152)	35 (23%)	16.5-30.5%	
Economic class* (n-6856)	Class I (Upper Class) (N = 839)	119 (14.2%)	11.8-16.7%	< 0.001*
	Class II (Upper Middle Class) (N= 1336)	188 (14.1%)	12.2-16.0%	
	Class III (Middle Class) (N= 2035)	392 (19.3%)	17.5-21.0%	
	Class IV (Lower Middle Class) (N= 1457)	287 (19.7%)	17.6-21.8%	
	Class V (Lower Class) (N= 1189)	182 (15.3%)	13.3-17.4%	
	Illiterate (N = 1876)	316 (16.8%)	15.1-18.6%	
Education status (n-6856)	Primary Education (N = 1321)	273 (20.7%)	18.5-22.9%	0.02*
	Middle School Education (N = 1803)	283 (15.7%)	14.0-17.4%	
	High School Education (N = 977)	155 (15.9%)	13.6-18.3%	
	Diploma (N = 200)	38 (19%)	13.8-25.1%	
	Graduate (N = 679)	103 (15.2%)	12.5-18.0%	

- With regards to proportion advised to get Oral cancer screening
 - More women advised than men.
 - No differences in locality.
 - More proportion advised among SC.
 - More proportion advised among lower middle class and middle class.
 - Those with primary education advised more

Table 27. Proportion ever screened for Oral Cancer

Variable	Stratum	N (%)	95% CI	P value
Overall	N = 6856	265 (3.87%)	3.4-4.3%	-
Gender	Men (N= 2575)	98 (3.81%)	3.1-4.6%	0.27
	Women (N= 4279)	167 (3.90%)	3.3-4.5%	
Locality (n-6856)	Rural (N = 4155)	162 (3.90%)	3.3-4.5%	0.87
	Urban (N=2701)	103 (3.81%)	3.1-4.6%	
Community groups (n-6856)	OC (N= 106)	1 (0.94%)	0.0-5.1%	0.05
	BC (N = 2416)	98 (4.06%)	3.3-4.9%	
	MBC (N = 1589)	76 (4.78%)	3.7-5.9%	
	SC (N=825)	32 (3.88%)	2.6-5.4%	
	ST (N= 220)	5 (2.27%)	0.7-5.2%	
	Not willing to reveal (N= 1700)	53 (3.12%)	2.3-4.0%	
Religion (n-6856)	Hindu (N = 6086)	239 (3.93%)	3.4-4.4%	0.02*
	Muslim (N= 254)	14 (5.51%)	3.0-9.0%	
	Christian (N = 357)	5 (1.40%)	0.4-3.2%	
	Others (N = 7)	0	0.0	
	Not willing to reveal (N = 152)	7 (4.61%)	1.8-9.2%	
Economic class* (n-6856)	Class I (Upper Class) (N = 839)	33 (3.93%)	2.7-5.4%	0.42
	Class II (Upper Middle Class) (N= 1336)	36 (2.69%)	1.8-3.7%	
	Class III (Middle Class) (N= 2035)	92 (4.52%)	3.6-5.5%	
	Class IV (Lower Middle Class) (N= 1457)	67 (4.60%)	3.5-5.8%	
	Class V (Lower Class) (N= 1189)	37 (3.11%)	2.2-4.2%	
	Illiterate (N = 1876)	74 (3.94%)	3.1-4.9%	
Education status (n-6856)	Primary Education (N = 1321)	55 (4.16%)	3.1-5.3%	0.24
	Middle School Education (N = 1803)	64 (3.55%)	2.7-4.5%	
	High School Education (N = 977)	30 (3.07%)	2.0-4.3%	
	Diploma (N = 200)	13 (6.50%)	3.5-10.8%	
	Graduate (N = 679)	29 (4.27%)	2.8-6.0%	

- With regards to proportion ever screened for Oral cancer
 - More women screened than men
 - More screening in rural areas than urban.
 - More proportion screened among MBC.
 - More proportion screened among lower middle class and middle class.
 - More proportion screened among diploma and graduates.

Table 28. Proportion advised to get Cervical Cancer Screening

Variable	Stratum	N (%)	95% CI	P value
Gender	Women (N = 4058)	1433 (35.3%)	33.8-36.8%	-
Locality (n-4058)	Rural (N= 2399)	844 (35.2%)	33.2-37.1%	0.83
	Urban (N = 1659)	589 (35.5%)	33.2-37.8%	
Community groups (n-4058)	OC (N = 61)	17 (27.9%)	17.1-40.8%	< 0.001*
	BC (N = 1486)	525 (35.3%)	32.9-37.8%	
	MBC (N = 947)	326 (34.4%)	31.4-37.5%	
	SC (N = 488)	174 (35.7%)	31.4-40.0%	
	ST (N = 127)	14 (11%)	6.1-17.8%	
	Not willing to reveal (N = 949)	377 (39.7%)	36.6-42.9%	
Religion (n-4058)	Hindu (N = 3592)	1261 (35.1%)	33.5-36.6%	0.07
	Muslim (N = 161)	50 (31.1%)	24.0-38.8%	
	Christian (N = 219)	82 (37.4%)	31.0-44.2%	
	Others (N = 2)	0	0	
	Not willing to reveal (N = 84)	40 (47.6%)	36.6-58.8%	
Economic class* (n-4058)	Class I (Upper Class) (N = 458)	181 (39.5%)	35.0-44.1%	< 0.001*
	Class II (Upper Middle Class) (N = 806)	299 (37.1%)	33.7-40.5%	
	Class III (Middle Class) (N = 1195)	456 (38.2%)	35.3-40.9%	
	Class IV (Lower Middle Class) (N = 871)	310 (35.6%)	32.4-38.8%	
	Class V (Lower Class) (N = 728)	187 (25.7%)	22.5-29.0%	
Education status (n-4058)	Illiterate (N = 1301)	384 (29.5%)	27.0-32.0%	< 0.001*
	Primary Education (N = 766)	272 (35.5%)	32.1-39.0%	
	Middle School Education (N = 1033)	375 (36.3%)	33.3-39.3%	
	High School Education (N = 537)	217 (40.4%)	36.2-44.7%	
	Diploma (N = 75)	36 (48%)	36.3-59.8%	
	Graduate (N = 346)	149 (43.1%)	37.7-48.4%	

- With regards to proportion advised to get Cervical cancer screening
 - No differences in locality.
 - More proportion advised among SC.
 - More proportion advised among upper class and middle class.
 - Those who were diploma and graduates were advised more

Table 29. Proportion ever had Cervical Cancer Screening

Variable	Stratum	N (%)	95% CI	P value
Gender	Women (N = 4058)	449 (11.06%)	10.1-12.0%	-
Locality (n-4058)	Rural (N= 2399)	242 (10.09%)	8.9-11.3%	0.01*
	Urban (N = 1659)	207 (12.48%)	10.9-14.1%	
Community groups (n-4058)	OC (N = 61)	2 (3.28%)	0.4-11.3%	0.18
	BC (N = 1486)	166 (11.17%)	9.6-12.8%	
	MBC (N = 947)	107 (11.30%)	9.3-13.4%	
	SC (N = 488)	57 (11.68%)	8.9-14.8%	
	ST (N = 127)	1 (0.79%)	0.0-4.3%	
	Not willing to reveal (N = 949)	116 (12.22%)	10.2-14.4%	
Religion (n-4058)	Hindu (N = 3592)	396 (11.02%)	10.0-12.0%	0.99
	Muslim (N = 161)	16 (9.94%)	5.7-15.6%	
	Christian (N = 219)	25 (11.42%)	7.5-16.3%	
	Others (N = 2)	0		
	Not willing to reveal (N = 84)	12 (14.29%)	7.6-23.6%	
Economic class* (n-4058)	Class I (Upper Class) (N = 458)	74 (16.16%)	12.9-19.8%	0.02*
	Class II (Upper Middle Class) (N = 806)	81 (10.05%)	8.0-12.3%	
	Class III (Middle Class) (N = 1195)	146 (12.22%)	10.4-14.2%	
	Class IV (Lower Middle Class) (N = 871)	90 (10.33%)	8.3-12.5%	
	Class V (Lower Class) (N = 728)	58 (7.97%)	6.1-10.1%	
	Illiterate (N = 1301)	127 (9.76%)	8.2-11.5%	
Education status (n-4058)	Primary Education (N = 766)	78 (10.18%)	8.1-12.5%	0.53
	Middle School Education (N = 1033)	109 (10.55%)	8.7-12.5%	
	High School Education (N = 537)	76 (14.15%)	11.3-17.3%	
	Diploma (N = 75)	10 (13.33%)	6.5-23.1%	
	Graduate (N = 346)	49 (14.16%)	10.6-18.2%	

- With regards to proportion ever screened for Cervical cancer
 - More screening in urban areas than rural.
 - More proportion screened among SC.
 - More proportion screened among Upper class and middle class.
 - More proportion screened among high school education and graduates.

Table 30. Proportion advised to get Breast Cancer Screening

Variable	Stratum	N (%)	95% CI	P value
Gender	Women (N = 4058)	1600 (39.4%)	37.9-40.9%	-
Locality (n-4058)	Rural (N= 2399)	975 (40.6%)	38.6-42.6%	0.05
	Urban (N = 1659)	625 (37.7%)	35.3-40.0%	
Community groups (n-4058)	OC (N = 61)	19 (31.1%)	19.9-44.2%	< 0.001*
	BC (N = 1486)	570 (38.4%)	35.8-40.8%	
	MBC (N = 947)	366 (38.6%)	35.5-41.8%	
	SC (N = 488)	201 (41.2%)	36.7-45.7%	
	ST (N = 127)	26 (20.5%)	13.8-28.5%	
	Not willing to reveal (N = 949)	418 (44%)	40.8-47.2%	
Religion (n-4058)	Hindu (N = 3592)	1419 (39.5%)	37.9-41.1%	0.06
	Muslim (N = 161)	50 (31.1%)	24.0-38.8%	
	Christian (N = 219)	91 (41.6%)	34.9-48.3%	
	Others (N = 2)	0		
	Not willing to reveal (N = 84)	40 (47.6%)	36.6-58.8%	
Economic class* (n-4058)	Class I (Upper Class) (N = 458)	200 (43.7%)	39.0-48.3%	< 0.001*
	Class II (Upper Middle Class) (N = 806)	335 (41.6%)	38.1-45.0%	
	Class III (Middle Class) (N = 1195)	502 (42%)	39.1-44.8%	
	Class IV (Lower Middle Class) (N = 871)	352 (40.4%)	37.1-43.7%	
	Class V (Lower Class) (N = 728)	211 (29%)	25.7-32.4%	
Education status (n-4058)	Illiterate (N = 1301)	443 (34.1%)	31.4-36.7%	< 0.001*
	Primary Education (N = 766)	306 (39.9%)	36.4-43.5%	
	Middle School Education (N = 1033)	411 (39.8%)	36.7-42.8%	
	High School Education (N = 537)	244 (45.4%)	41.1-49.7%	
	Diploma (N = 75)	37 (49.3%)	37.5-61.1%	
	Graduate (N = 346)	159 (46%)	40.6-51.3%	

- With regards to proportion advised to get Breast cancer screening
 - More proportion advised in rural than urban.
 - More proportion advised among SC.
 - More proportion advised among upper class and middle class.
 - Those who were diploma and graduates were advised more.

Table 31. Proportion ever had Breast Cancer Screening

Variable	Stratum	N (%)	95% CI	P value
Gender	Women (N = 4058)	578 (14.24%)	13.1-15.3%	-
Locality (n-4058)	Rural (N= 2399)	339 (14.13%)	12.7-15.5%	0.15
	Urban (N = 1659)	239 (14.41%)	12.7-16.1%	
Community groups (n-4058)	OC (N = 61)	5 (8.20%)	2.7-18.1%	0.42
	BC (N = 1486)	193 (12.99%)	11.3-14.8%	
	MBC (N = 947)	129 (13.62%)	11.5-15.9%	
	SC (N = 488)	74 (15.16%)	12.1-18.6%	
	ST (N = 127)	10 (7.87%)	3.8-14.0%	
	Not willing to reveal (N = 949)	167 (17.60%)	15.2-20.1%	
Religion (n-4058)	Hindu (N = 3592)	511 (14.23%)	13.1-15.4%	0.14
	Muslim (N = 161)	16 (9.94%)	5.7-15.6%	
	Christian (N = 219)	30 (13.70%)	9.4-18.9%	
	Others (N = 2)	0		
	Not willing to reveal (N = 84)	21 (25.00%)	16.1-35.6%	
Economic class* (n-4058)	Class I (Upper Class) (N = 458)	105 (22.93%)	19.1-27.0%	< 0.001*
	Class II (Upper Middle Class) (N = 806)	109 (13.52%)	11.2-16.0%	
	Class III (Middle Class) (N = 1195)	170 (14.23%)	12.2-16.3%	
	Class IV (Lower Middle Class) (N = 871)	120 (13.78%)	11.5-16.2%	
	Class V (Lower Class) (N = 728)	74 (10.16%)	8.0-12.5%	
Education status (n-4058)	Illiterate (N = 1301)	156 (11.99%)	10.2-13.8%	0.08
	Primary Education (N = 766)	99 (12.92%)	10.6-15.5%	
	Middle School Education (N = 1033)	152 (14.71%)	12.6-17.0%	
	High School Education (N = 537)	83 (15.46%)	12.5-18.8%	
	Diploma (N = 75)	15 (20.00%)	11.6-30.8%	
	Graduate (N = 346)	73 (21.10%)	16.9-25.7%	

- With regards to proportion ever screened for Breast cancer
 - More screening in urban areas than rural.
 - More proportion screened among SC.
 - More proportion screened among Upper class and middle class.
 - More proportion screened among graduates.

Table 32. Proportion covered under Palliative Care through MTM-field staff

Variable	Stratum	N (%)	95% CI	P value
Overall	N = 319	126 (39.50%)	34.1-45.1%	-
Gender (N = 319)	Men (N = 128)	49 (38.28%)	29.8-47.2%	0.98
	Women (N = 191)	75 (39.27%)	32.3-46.5%	
Locality (N = 319)	Rural (N = 192)	95 (49.48%)	42.2-56.7%	0.001*
	Urban (N = 127)	31 (24.41%)	17.2-32.8%	
Community groups (N = 319)	OC (N = 5)	2 (40.00%)	5.2-85.3%	0.17
	BC (N = 117)	46 (39.32%)	30.4-48.7%	
	MBC (N = 71)	23 (32.39%)	21.7-44.5%	
	SC (N = 33)	13 (39.39%)	22.9-57.8%	
	ST (N = 12)	9 (75%)	42.8-94.5%	
	Not willing to reveal (N = 81)	32 (39.51%)	28.8-50.9%	
Religion (N = 319)	Hindu (N = 278)	116 (41.73%)	35.8-47.7%	0.001*
	Muslim (N = 11)	7 (63.64%)	30.7-89.0%	
	Christian (N = 18)	2 (11.11%)	1.3-34.7%	
	Others (N = 0)	0		
	Not willing to reveal (N = 12)	1 (8.33%)	0.2-38.4%	
Economic class* (N = 319)	Class I (Upper Class) (N = 46)	9 (19.57%)	9.3-33.9%	0.06
	Class II (Upper Middle Class) (N = 55)	18 (32.73%)	20.6-46.7%	
	Class III (Middle Class) (N = 89)	40 (44.94%)	34.3-55.8%	
	Class IV (Lower Middle Class) (N = 69)	32 (46.38%)	34.2-58.8%	
	Class V (Lower Class) (N = 60)	27 (45.00%)	32.1-58.3%	
Education status (N = 319)	Illiterate (N = 91)	43 (47.25%)	36.6-58.0%	0.002*
	Primary Education (N = 85)	43 (50.59%)	39.5-61.6%	
	Middle School Education (N = 69)	19 (27.54%)	17.4-39.6%	
	High School Education (N = 39)	12 (30.77%)	17.0-47.5%	
	Diploma (N = 8)	4 (50.00%)	15.7-84.3%	
	Graduate (N = 27)	5 (18.52%)	6.3-38.0%	

- With regards to proportion receiving palliative care through MTM field staff
 - More coverage among women than men.
 - More coverage in rural areas than urban.
 - More coverage among ST.
 - More coverage among lower middle class and lower class.
 - More coverage among those with primary education.

Table 33. Proportion receiving Physiotherapy services through MTM-field staff

Variable	Stratum	N (%)	95% CI	P value
Overall	N = 320	153 (47.81%)	42.2-53.4%	-
Gender (N = 320)	Men (N = 135)	61 (45.19%)	36.6-53.9%	0.03*
	Women (N = 185)	92 (49.73%)	42.3-57.1%	
Locality (N = 320)	Rural (N = 191)	120 (62.83%)	55.5-69.6%	< 0.001*
	Urban (N = 129)	33 (25.58%)	18.3-34.0%	
Community groups (N = 320)	OC (N = 7)	5 (71.43%)	29.0-96.3%	< 0.001*
	BC (N = 128)	67 (52.34%)	43.3-61.2%	
	MBC (N = 73)	38 (52.05%)	40.0-63.9%	
	SC (N = 28)	17 (60.71%)	40.5-78.5%	
	ST (N = 10)	7 (70.00%)	34.7-93.3%	
	Not willing to reveal (N = 74)	19 (25.68%)	16.2-37.1%	
Religion (N = 320)	Hindu (N = 274)	145 (52.92%)	46.8-58.9%	< 0.001*
	Muslim (N = 20)	7 (35.00%)	15.3-59.2%	
	Christian (N = 11)	0		
	Others (N = 1)	0		
	Not willing to reveal (N = 14)	1 (7.14%)	0.1-33.8%	
Economic class* (N = 320)	Class I (Upper Class) (N = 59)	14 (23.73%)	13.6-36.5%	< 0.001*
	Class II (Upper Middle Class) (N = 75)	32 (42.67%)	31.3-54.6%	
	Class III (Middle Class) (N = 81)	36 (44.44%)	33.4-55.9%	
	Class IV (Lower Middle Class) (N = 57)	32 (56.14%)	42.3-69.2%	
	Class V (Lower Class) (N = 48)	39 (81.25%)	67.3-91.0%	
Education status (N = 320)	Illiterate (N = 78)	53 (67.95%)	56.4-78.0%	< 0.001*
	Primary Education (N = 64)	31 (48.44%)	35.7-61.2%	
	Middle School Education (N = 76)	34 (44.74%)	33.3-56.5%	
	High School Education (N = 46)	22 (47.83%)	32.8-63.0%	
	Diploma (N = 13)	5 (38.46%)	13.8-68.4%	
	Graduate (N = 43)	8 (18.60%)	8.3-33.4%	

- With regards to proportion covered receiving physiotherapy services through MTM field staff
 - More coverage among women than men.
 - More coverage in rural areas than urban.
 - More coverage among ST.
 - More coverage among lower class.
 - More coverage among illiterates.

Table 34. Proportion receiving Dialysis services through MTM-field functionary

Variable	Stratum	N (%)	95% CI	P value
Overall	N = 43	22 (51.16%)	35.4-66.6%	-
Gender (N = 43)	Men (N = 23)	10 (43.48%)	23.1-65.5%	0.25
	Women (N = 20)	12 (60%)	36.0-80.8%	
Locality (N = 43)	Rural (N = 24)	11 (45.83%)	25.5-67.1%	0.70
	Urban (N = 19)	11 (57.89%)	33.5-79.7%	
Community groups (N = 43)	OC (N = 0)	0	0	0.02*
	BC (N = 20)	15 (75%)	50.9-91.3%	
	MBC (N = 7)	2 (28.6%)	3.6-70.9%	
	SC (N = 1)	1 (100%)	2.5-100.0%	
	ST (N = 3)	1 (33.3%)	0.8-90.5%	
	Not willing to reveal (N = 12)	3 (25%)	5.4-57.1%	
Religion (N = 43)	Hindu (N = 38)	20 (52.6%)	35.8-69.0%	0.53
	Muslim (N = 3)	1 (33.3%)	0.8-90.5%	
	Christian (N = 2)	1 (50%)	1.2-98.7%	
	Others (N = 0)	0	0	
	Not willing to reveal (N = 1)	0	0	
Economic class* (N = 43)	Class I (Upper Class) (N = 4)	3 (75%)	19.4-99.3%	0.40
	Class II (Upper Middle Class) (N = 13)	7 (53.8%)	25.1-80.7%	
	Class III (Middle Class) (N = 10)	4 (40.00%)	12.1-73.7%	
	Class IV (Lower Middle Class) (N = 10)	7 (70.00%)	34.7-93.3%	
	Class V (Lower Class) (N = 7)	1 (14.3%)	0.3-57.8%	
Education status (N = 43)	Illiterate (N = 9)	5 (55.6%)	21.2-86.3%	0.46
	Primary Education (N = 12)	6 (50%)	21.0-78.9%	
	Middle School Education (N = 8)	3 (37.5%)	8.5-75.5%	
	High School Education (N = 5)	3 (60%)	14.6-94.7%	
	Diploma (N = 2)	2 (100%)	15.8-100.0%	
	Graduate (N = 7)	3 (42.9%)	9.9-81.5%	

- With regards to proportion covered receiving dialysis services through MTM field staff
 - More coverage among women than men.
 - More coverage in urban areas than rural.
 - More coverage among SC.
 - More coverage among Upper class.
 - More coverage among diploma holders and those with high school education.

Figure 3. Association between receipt of drugs through field staff and diabetes /hypertension control

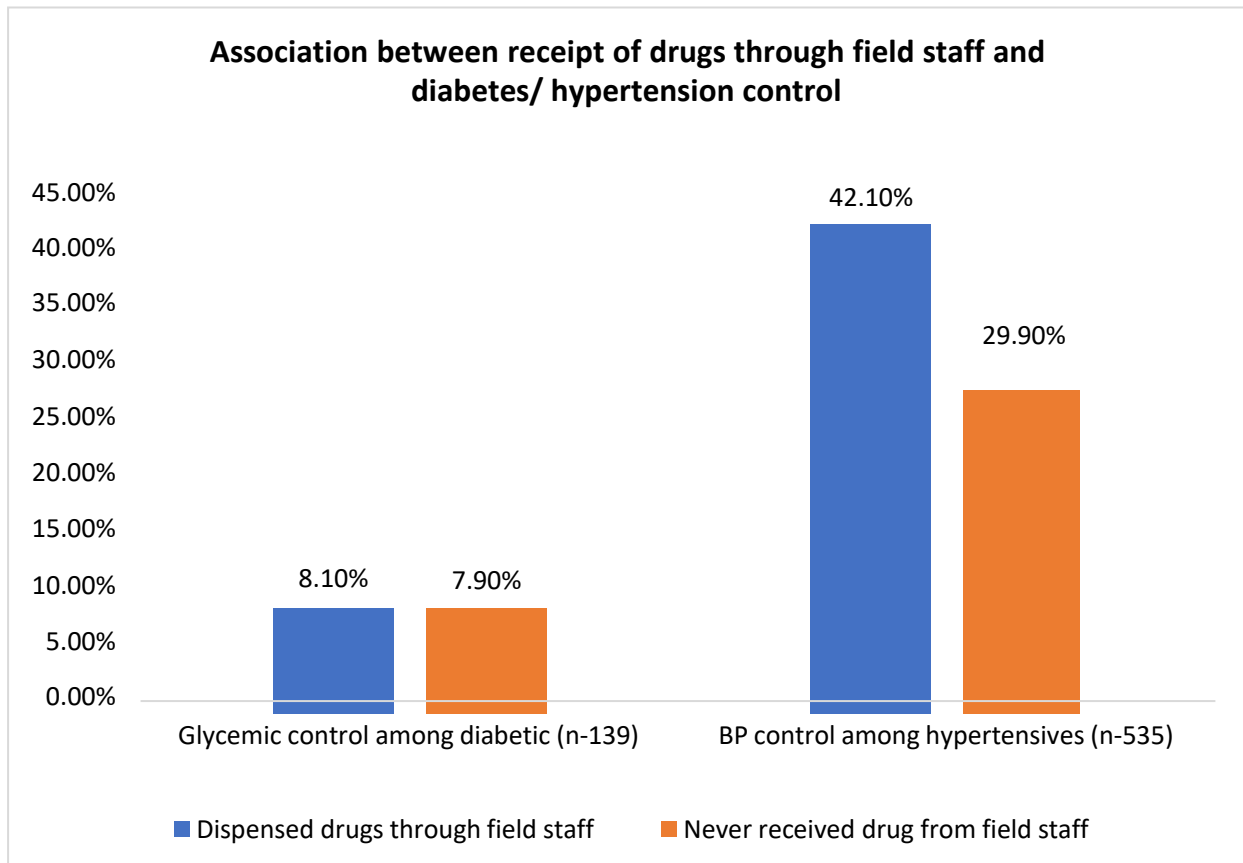


Figure 4. Proportion who preferred home services under MTM

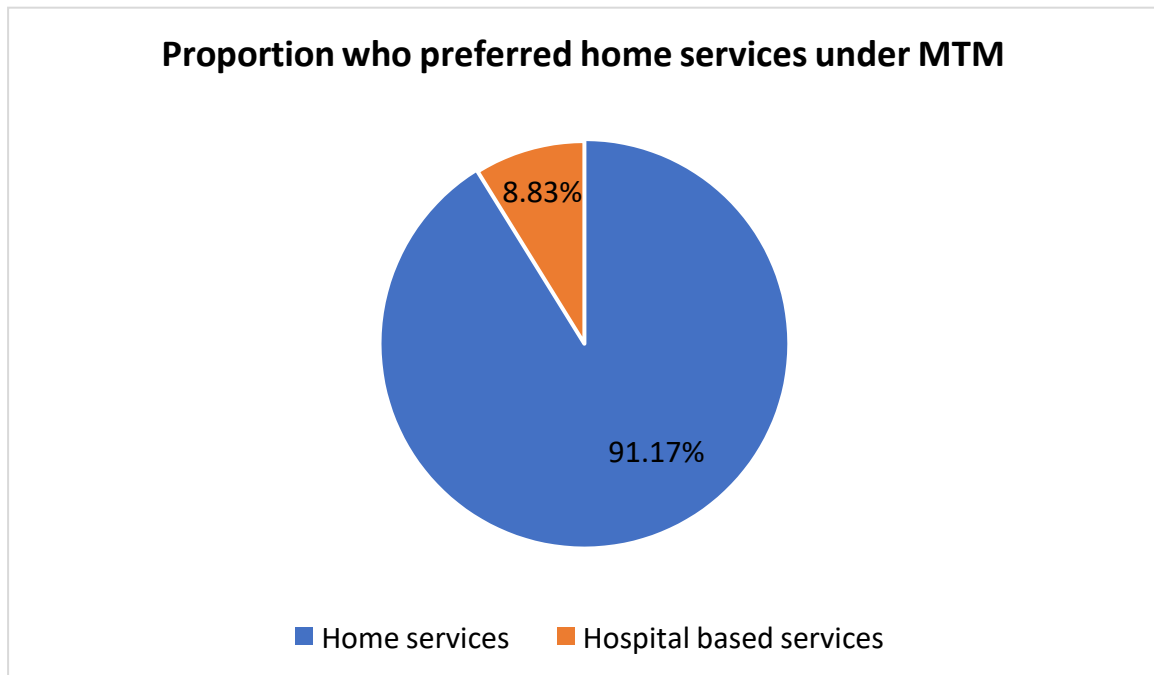
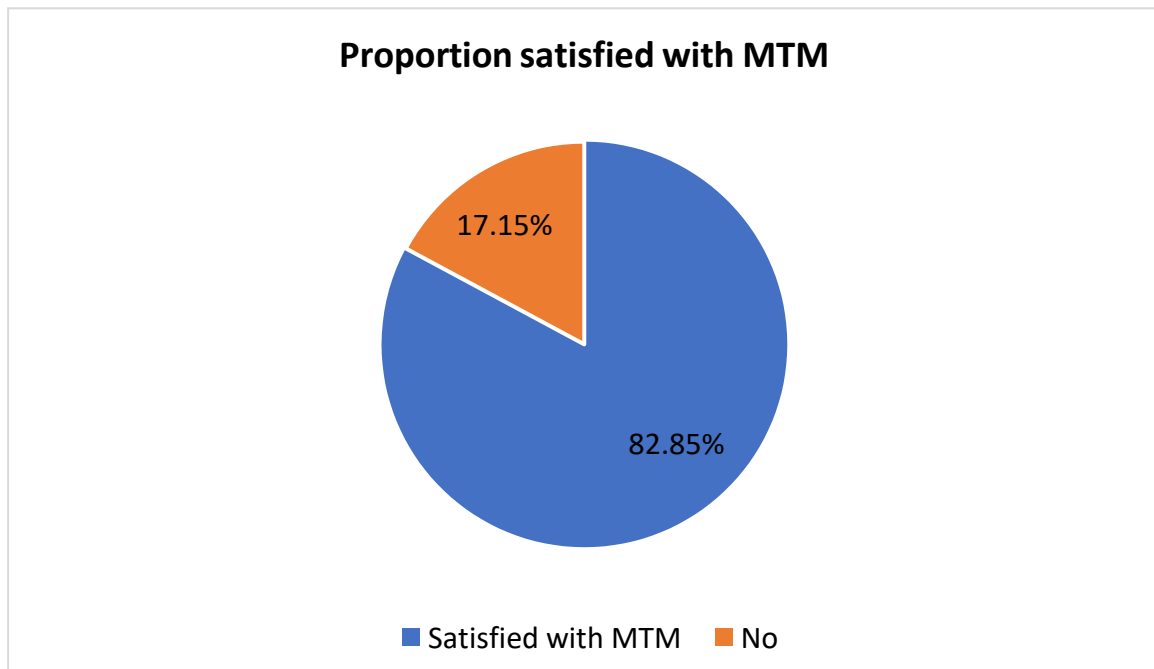


Figure 5. Proportion satisfied with MTM



SUMMARY OF KEY FINDINGS

Among the study participants, 4155 (60%) were from rural areas and 2701 (40%) belonged to urban localities. Gender distribution was 2575 (37.5%) men, and 4279(62.4%) were women. Based on Community, 24.8% did not want to tell their community. Among those who revealed, 2416 (35.2%) belonged to BC, 1589 (23.2%) belonged to MBC ,825 (12%) belonged to the SC and 220 (3.2%) belonged to the ST

Almost 78.84% of the study participants were aware of MTM and 73 % were ever visited by WHV. Among those visited by WHV, 90.48% were briefed about MTM services by WHV. While there was no gender difference, the awareness on MTM and proportion of people visited by a WHV was significantly lower in urban areas compared to rural areas. Similarly, awareness on MTM was 79% in BC , 84.7% in MBC , 83.1% in SC and 87.27% in OC . Significantly higher proportion of SC and ST community people were visited by a WHV and were briefed about MTM services compared to non-SC/STpeople.

Screening for Hypertension: Almost 81.25% of the whole study population have ever been screened for Hypertension, among them 93.8% had screened themselves for Hypertension in the last year. Among those screened in the last year, 73.6% was screened through MTM. 82.9% were screened in rural areas and 61.5% in urban areas. 67.1% were screened in BC, 78.1% in MBC, 87.5% in SC and 92.6% in ST were screened by MTM.75.6% in Hindus, 69.2% in Christians and 51.5% in Muslims were screened by MTM. Specifically, almost 50% were screened by field workers at doorstep under MTM.

A higher proportion of women were screened compared to men (82.6% vs 79%). Screening through MTM in the last year was also higher among women (75.1% vs 71.2%). Under MTM, screening by field workers at the doorstep was high among men compared to women (72.5% vs 68.3%)

There was an urban-rural difference in overall hypertension screening coverage, with a higher coverage rate in rural areas (82.5%) compared to urban areas (79.4%). The screening coverage through MTM for Hypertension in urban areas was 61.5% vs 82.9% in rural areas. 66.3% were screened by MTM field workers in urban localities compared to 73% in rural areas.

Based on community there was no significant difference in overall screening coverage. However, **screening coverage through MTM services was significantly higher among SC**

and ST categories compared to non-SC/ST categories. Specifically, the ST population (91.4%) had a higher proportion screened through MTM field workers compared to SC and OC,BC &MBC population (64.7% & 49.1%). MTM screening through institutions was significantly higher among SC (23%) and lowest among ST (7.4%) compared to OC,BC &MBC (17.8%).

Proportion of Hypertension

Among the study population, 22% were hypertensive. There was no gender difference, but urban population had a higher proportion of hypertensives compared to rural areas (25% vs 20%). People belonging to OC,BC &MBC had a higher proportion of HTN compared to SC (23%vs 18.3%).

Among the hypertensives, 1/5th were diagnosed in the last 1 year and 96% reported to be on treatment. There was no significant difference in the treatment coverage based on gender, community and locality. Among the hypertensives, 2/3rd were receiving treatment under MTM and specifically, 41% of the Hypertensives had received treatment at their doorsteps through field workers. There was a significant difference in treatment received through MTM based on locality. Only 48% of hypertensives in urban areas received treatment through MTM, compared to 74% in rural areas. Similarly, only 30.1% of hypertensives in urban areas had been dispensed drugs through field workers compared to 57.6% in rural areas. Among different community, 3/4th of the hypertensives belonging to SC and ST had received treatment through MTM compared to only 60% among OC, BC & MBCnon-SC/ST category.

Follow up and Control

Among the hypertensives, 90% reported compliance to drugs in the last 1 week and 70% had recorded their blood pressure at least once in the past 3 months. Based on the latest recorded BP within the last 3 months, 35.4% of the hypertensives had their BP under control. There was no gender, community and locality-based difference.

Diabetes Mellitus

Among the study participants, 80% had screened for DM. Among those screened, 93% had their screening done in the last one year. **69 percent of those screened in last one year, were screened through MTM.** Specifically, 47% were screened by field workers at their doorstep and 21% through institutions. **A higher proportion of women had screened for DM compared to men (81% vs 77%).** There was no urban rural difference. People belonging to

ST category had a significantly higher proportion screened for DM compared to all other community..

Among those screened through MTM in the last one year, there was a significant gender difference with preponderance towards women. Similarly, a higher proportion of women (71.9%) were visited by field workers under MTM compared to men (65.5%).

Rural counterparts (77%) had a higher proportion screened for DM through MTM compared to urban (53%) which was also reflected in screening through field workers (70.8% vs 66.9%) and institutional screening (22.4% vs 17.5%).

With respect to caste, 63.4% among BC, 72.7% among MBC, 79.9% among SC and 84.9% among ST were screened for DM through MTM. Specifically, 71.7% among BC, 73.8% among MBC, 64.9% among SC and 85.5% among ST were screened for DM by field workers in MTM.

Such difference was also observed in proportion screened through field workers with the highest reported among ST category 85.5%. Institutional screening for DM under MTM was highest among SC (18.22%) and the lowest among ST community. (12.29%)

The overall proportion of people who reported to be a diabetic was 21%, with men reporting higher compared to women (22.5% vs 19.6%). Similarly urban had higher proportion of DM reported compared to rural (25.2% versus 17.8%). There was caste difference in the DM proportion with highest among OC,BC&MBC (22%)and lowest in ST(9.55%).

Among those diagnosed with DM, 16.6% were diagnosed in the last one year and 97% reported to be on treatment. Fifty five percent of those with DM received treatment under MTM and 36% got their Medicines dispensed through Field workers. A significantly higher proportion of rural population (65.2%) with DM were covered under treatment by MTM compared to urban (42.9%). Similarly 45.3% of those with DM in rural areas were dispensed drugs by field workers whereas only 25.9% in urban.

With regards to community category, OC,BC &MBC had a significantly lower proportion receiving treatment through MTM compared to SC and ST. Similarly, 42.9% of ST people with DM were receiving drugs dispensed through Field workers.

Follow up and control

Among the diabetic, 66% had their blood sugars tested in last 3 months and 9.8% of the diabetics had their sugar levels under control. There was no difference based on gender, community and locality.

Cancer screening

Cancer screening uptake was lower overall. Only 4%, had ever undergone screening for oral cancer. 11.06% of women had ever undergone screening for cervical cancer and 14.24% of women had ever undergone screening for breast cancer. There was no rural-urban difference and gender difference (oral cancer). However, ST had the lowest screening coverage for oral cancer and cervical cancer, compared to other communities.

Counselling services

Under MTM , counselling for cancer screening should be provided as a service. However, only 17%, 35% and 39% had ever received counselling regarding screening for oral, cervix and breast cancer respectively. While there was no difference in this proportion, based on locality, difference based on Communities was obvious. Counselling services were reported to be lowest among ST category compared to others.

Coverage based on Occupation (Appendix)

87.3% of casual laborers, 73% of non-casual laborers and 69.15% of unemployed/ homemakers were screened for HTN by MTM. 81.8% of casual laborers, 77.1% of non-casual laborers and 68.9% of unemployed/ homemaker were diagnosed as HTN by MTM. Similarly, 77.3% of casual laborers, 63.5% of non-casual laborers and 58.8% of unemployed/ homemaker were on treatment for HTN through MTM.

80.3% of casual laborers, 67.3% of non-casual laborers and 63.3% of unemployed/ homemakers were screened for DM by MTM. 85.7% of casual laborers, 65.9% of non-casual laborers and 57% of unemployed/ homemaker were diagnosed as DM by MTM. Similarly, 71.7% of casual laborers, 54.9% of non-casual laborers and 50.6% of unemployed/ homemaker were on treatment for DM through MTM.

28.6% of casual laborers, 19.9% of non-casual laborers and 22.3% of unemployed/ homemaker were screened for oral cancer. 14.4% of casual laborers, 10.0% of non-casual laborers and 10.7% of unemployed/ homemaker were screened for cervical cancer. Similarly, 17.7% of casual laborers, 13.5% of non-casual laborers and 13.7% of unemployed/ homemaker were screened for breast cancer.

93.3% of casual laborers, 67.1% of non-casual laborers and 56.6% of unemployed/ homemaker were covered under MTM for palliative care. 76.6% of casual laborers, 68% of non-casual laborers and 63% of unemployed/ homemaker were covered under MTM for physiotherapy. 83.3% of casual laborers, 86.9% of non-casual laborers and 73.3% of unemployed/ homemaker were covered under MTM for dialysis services.

94% of casual laborers, 89.7% of non-casual laborers and 91.4% of unemployed/ homemaker preferred home services under MTM. 98.4% of casual laborers, 97.1% of non-casual laborers and 97.4% of unemployed/ homemaker were satisfied with MTM services.

RECOMMENDATIONS

Overall

- Diabetes and hypertension are the most widely covered NCDs. Cancer screening coverage, on the other hand, is extremely low. Cancer screening counselling is also appallingly inadequate.
- Cancer screening demand should be increased by raising awareness.
- Counselling for cancer screening in the field should be given priority.
- Plans for educating beneficiaries about cancer screening should be developed.
- WHV may be adequately trained in cancer screening counselling.

Gender Difference

- Women are better covered than men across the board, but men have a higher disease burden (DM and HTN). This could be because the current strategy of house-to-house visits by Women Health Volunteers focuses primarily on women, who are more available at home than men.
- Men-focused policies should be put into place. NCD workplace screening needs to be improved.

Locality Difference

- The coverage of services in rural areas is almost universal. However, the coverage through MTM in urban areas is subpar.
- This might be as a result of the fact that there is already a strong infrastructure for public health in rural areas and that there is an effective system in place for outreach services there.
- Urban infrastructure still needs to be improved. Urban areas lack the necessary infrastructure or human resources to offer outreach services.
- The number of people to be covered by field staff is very large, limiting their ability to provide quality services.
- Urban gets their NCD care services through non MTM resources. Further exploratory studies should be undertaken to their preferred place to get services and the reasons for such preferences. Based on which, urban specific strategies should be devised to ensure coverage through MTM.

To further strengthen service provision through MTM, continuum of care should be strengthened.

To enable continuum of care, following areas should be targeted.

- Women Health Volunteers are the key personnel in delivering NCD care services. The importance of this role is well established. Efforts should be taken to ensure their service delivery without interruption. Decrease attrition among Women Health Volunteers by providing honorarium based on their performance.
- After screening, continuum of care through hospitals should be ensured. Referral linkage using digital platforms to be strengthened.

Conclusion

MTM has been a pro-poor service, reaching out to the most vulnerable groups such as women, rural areas, and the SC/ST category. MTM services are essentially a successful strategy for achieving universal health care.

Diabetes and hypertension screening and treatment coverage has reached the greatest number of people among the various services provided. This suggests that de-professionalization of health care through the involvement of Women Health Volunteers has had a significant impact in reaching out to the most vulnerable population. While the MTM enables universal NCD coverage, the thrust areas include reaching out to urban areas and developing strategies for improving overall cancer screening.

APPENDIX

Table 1. Coverage based on Occupation

Services	Casual laborer (n = 1087)	Non casual laborer (n = 2875)	Unemployed/ homemaker (n = 2894)	P value
Proportion Aware of MTM (n-6856)	902 (83%)	2290 (79.7%)	2213 (76.5%)	0.001*
Proportion ever visited by MTM field functionary (WHV) (n-6856)	869 (79.9%)	2103 (73.1%)	2016 (69.7%)	0.001*
Proportion briefed about MTM by WHV(n-4988)	621 (71.4%)	1446 (68.7%)	1365 (67.7%)	0.001*
Proportion ever screened for DM (n-6856)	870 (80%)	2220 (77.2%)	2358 (81.5%)	0.001*
Proportion screened for DM in the last 1 year (n-5448)	813 / 870 (93.4%)	2067 / 2220 (93.1%)	2185 / 2358 (92.7%)	0.70
Proportion screened for DM through MTM (n-5448)	699 / 870 (80.3%)	1494 / 2220 (67.3%)	1492 / 2358 (63.3%)	0.001*
Proportion screened for DM through MTM by field functionaries (n-3685)	484 / 699 (69.2%)	1056 / 1494 (70.7%)	1026 / 1492 (68.8%)	0.51
Proportion screened for DM through MTM institutions. (n-3685)	215 / 699 (30.8%)	438 / 1494 (29.3%)	466 / 1492 (31.2%)	0.51
Proportion ever diagnosed with DM (n-6856)	152 (14.0%)	556 (19.3%)	711 (24.6%)	0.001*
New diagnosis for DM in the last one year (n-1419)	35 / 152 (2.5%)	94 / 556 (16.9%)	107 / 711 (15.1%)	0.80
New diagnosis for DM in the last one year through MTM (n-236)	30 / 35 (85.7%)	62 / 94 (65.9%)	61 / 107 (57.0%)	0.01*
Proportion on treatment for DM (n-1419)	146 / 152 (96.1%)	536 / 556 (97.3%)	700 / 711 (98.5%)	0.001*
Proportion on treatment for DM through MTM (n-1419)	109 / 152 (71.7%)	305 / 556 (54.9%)	360 / 711 (50.6%)	0.03*
Proportion of DM dispensed medicines through WHV for DM (n-1419)	64 / 152 (42.1%)	204 / 556 (36.7%)	243 / 711 (34.2%)	0.001*
Proportion of DM who have checked their blood glucose level in the last 3 months(n-1419)	114 / 152 (75.0%)	344 / 556 (61.9%)	482 / 711 (67.8%)	0.001*
Proportion of DM with good glycemic control (n-1419)	25 / 152 (16.4%)	41 / 556 (7.4%)	73 / 711 (10.3%)	0.001*
Proportion of DM patients who shifted from private to public (n-331)	8 / 33 (24.2%)	52 / 141 (36.9%)	37 / 157 (23.6%)	0.13
Proportion ever screened for HTN (n-6856)	892 (82.1%)	2279 (79.3%)	2399 (82.9%)	0.002*
Proportion screened for HTN in the last 1 year (n-6856)	840 (94.2%)	2132(93.5%)	2253(93.9%)	0.77
Proportion screened for HTN through MTM (n-5225)	733(87.3%)	1557(73%)	1557(69.15)	0.000*
Proportion screened for HTN through MTM by field functionaries (n-3847)	532(72.6%)	1123(72.1%)	1077(69.2%)	0.112
Proportion screened for HTN through MTM institutions. (n-3847)	201(27.4%)	434(27.9%)	480(30.8%)	0.112
Proportion ever diagnosed with HTN (n-6856)	181 (20.3%)	529 (23.2%)	801 (33.4%)	0.000*

New diagnosis for HTN in the last one year (n-1511)	44 (24.3%)	118 (22.3%)	132 (16.5%)	0.007*
New diagnosis for HTN in the last one year through MTM (n-289)	36(81.8%)	91(77.1%)	91(68.9%)	0.152
Proportion on treatment for HTN (n-1511)	172(95%)	503(95.1%)	779(97.3%)	0.084
Proportion on treatment for HTN through MTM (n-1511)	140(77.3%)	336(63.5%)	471(58.8%)	0.000*
Proportion of HTN dispensed medicines through WHV (n-1511)	93(51.4%)	219(41.4%)	312(39%)	0.009*
Proportion of HTN who have checked their blood pressure level in the last 3 months (n-1511)	139(76.8%)	348(65.8%)	563(70.3%)	0.016*
Proportion of HTN with good blood pressure control (n-1511)	79(43.6%)	169(31.9%)	287(35.8%)	0.017*
Proportion of HTN patients who shifted from private to public (n-292)	13(29.5%)	32(23%)	31(28.4%)	0.531
Services related to Cancer				
Proportion advised to get Oral Cancer Screening (n-6856)	252 (23.2%)	473 (16.5%)	443 (15.3%)	0.001*
Proportion ever screened for Oral Cancer (n-6856)	72 (28.6%)	94 (19.9%)	99 (22.3%)	0.02*
Proportion advised to get Cervical Cancer Screening (n-4058)	214 / 562 (38.0%)	396 / 1088 (36.3%)	823 / 2408 (34.1%)	0.15
Proportion ever had Cervical Cancer Screening (n-4058)	81 / 562 (14.4%)	109 / 1088 (10.0%)	259 / 2408 (10.7%)	0.03*
Proportion advised to get Breast Cancer Screening (n-4058)	236 / 562 (41.9%)	446 / 1088 (40.9%)	918 / 2408 (38.1%)	0.11
Proportion ever had Breast Cancer Screening (n-4058)	100 / 562 (17.7%)	147 / 1088 (13.5%)	331 / 2408 (13.7%)	0.05
Other services				
Proportion covered under Palliative Care through MTM (n-319)	28 / 30 (93.3%)	98 / 146 (67.1%)	81 / 143 (56.6%)	0.009*
Proportion covered under Palliative Care through MTM-field staff (n-319)	16 / 30 (53.3%)	59 / 146 (40.4%)	51 / 143 (35.6%)	0.009*
Proportion receiving Physiotherapy services through MTM (n-320)	23 / 30 (76.6%)	96 / 141 (68.0%)	94 / 149 (63.0%)	0.004*
Proportion receiving Physiotherapy services through MTM-field staff (n-320)	16 / 30 (53.3%)	59 / 141 (41.8%)	78 / 149 (52.3%)	0.004*
Proportion receiving Dialysis services through MTM (n-43)	5 / 6 (83.3%)	20 / 23 (86.9%)	11 / 15 (73.3%)	0.53
Proportion receiving Dialysis services through MTM-field staff (n-43)	4 / 6 (66.6%)	11 / 23 (47.8%)	7 / 15 (46.6%)	0.53
Proportion who preferred home services under MTM (n-5184)	827 / 879 (94.0%)	1957 / 2181 (89.7%)	1942 / 2124 (91.4%)	0.001*
Proportion satisfied with MTM (n-5184)	865 / 879 (98.4%)	2119 / 2181 (97.1%)	2070 / 2124 (97.4%)	0.001*

Evaluation of Non-Communicable Disease Cascade Care under *Makkalai Thedi Maruthuvam* in Tamil Nadu

The *Makkalai Thedi Maruthuvam* scheme stands as a flagship initiative of the Tamil Nadu government, designed to deliver comprehensive Home-Based Health Care Services. The program is strategically crafted to ensure a seamless continuum of care, with a focus on a field-level approach, offering home-based health care services with a special emphasis on **non-communicable diseases (NCD)**.

Over a year has passed since the expansion of MTM services to encompass the entire state of Tamil Nadu. Thus, it becomes imperative to ascertain the extent of coverage provided by the MTM scheme across the region. This study has been commissioned by the State Planning Commission with the explicit aim of evaluating the reach and effectiveness of MTM services in Tamil Nadu.



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